



Cyclone Risk Assessment of Shyamnagar Sub District, Bangladesh: Agrarian Transformation and Health Risk Perception

Submitted To

Department of Coastal Studies and Disaster Management
University of Barishal

Supervised By

Dr. Hafiz Ashraful Haque
Associate Professor, Chairman
Department of Coastal Studies and Disaster Management
University of Barishal

Submitted By

Md. Osman Goni
Exam Roll: CDM 013/4
Registration Number: 191-002-20
Session: 2019-20
Department of Coastal Studies and Disaster Management
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Declaration

I affirm that the work presented in this project is the outcome of my own research and investigation. To the best of my knowledge, this project has not been submitted in full to any other university or institution for the award of any degree or qualification.

Md. Osman Goni

Date

Name and Signature

Certificate of Supervisor

I am pleased to certify that the research work presented in this project, titled " Cyclone Risk Assessment of Shyamnagar Sub District, Bangladesh: Agrarian Transformation and Health Risk Perception ", was carried out by Md. Osman Goni in partial fulfillment of the requirements for the Bachelor of Science (BSc) degree. I had the privilege of supervising this research throughout its entirety, and I can confirm that the work reflects genuine intellectual effort and academic dedication on the part of the researcher. Md. Osman Goni approached this study with curiosity and commitment, engaging thoughtfully with the subject matter at every stage of the research process. Throughout the course of this project, the researcher has duly observed the academic rules, guidelines, and ethical standards set forth by the University of Barishal. The findings and conclusions presented here are drawn entirely from his own investigation and analysis, and to the best of my knowledge, this work has not been previously submitted to any other university or institution for the purpose of obtaining a degree or for publication. I believe this study makes a meaningful contribution to our understanding of coastal vulnerability, cyclone risk, and land use dynamics in the context of Bangladesh. In my honest assessment, this project is well worthy of consideration for the award of the Bachelor of Science (BSc) degree.

Supervisor

Dr. Hafiz Ashraful Haque

Associate Professor, Chairman

Department of Coastal Studies and Disaster Management

University of Barishal

Date

List of Abbreviations

Abbreviation	Full Form
AHP	Analytical Hierarchy Process
BMD	Bangladesh Meteorological Department
BBS	Bangladesh Bureau of Statistics
CI	Consistency Index
CR	Consistency Ratio
CVI	Composite Vulnerability Index
DDM	Department of Disaster Management
DEM	Digital Elevation Model
DRR	Disaster Risk Reduction
EI	Exposure Index
ESA	European Space Agency
FIR	Flood Inundation Risk
GIS	Geographic Information System
IPCC	Intergovernmental Panel on Climate Change
JPL	Jet Propulsion Laboratory
LGED	Local Government Engineering Department
LULC	Land Use Land Cover
MCDA	Multi-Criteria Decision Analysis
NASA	National Aeronautics and Space Administration
NASADEM	NASA Digital Elevation Model
NDVI	Normalized Difference Vegetation Index
RI	Random Index
RS	Remote Sensing
SFDRR	Sendai Framework for Disaster Risk Reduction
SRTM	Shuttle Radar Topography Mission
USD	United States Dollar

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Abstract:

Coastal regions of Bangladesh are highly vulnerable to cyclones, and climate change is expected to make these storms more frequent and severe in the coming decades. Despite growing exposure, spatially explicit risk assessments integrating hazard, vulnerability, and mitigation capacity remain limited for sub-district scales. This study develops a comprehensive cyclone risk map for Shyamnagar Upazila using the Analytic Hierarchy Process (AHP) applied to a weighted overlay of 15 spatial criteria across hazard (incorporating a novel Storm Surge Composite Vulnerability Index), vulnerability, and mitigation capacity components within a GIS environment. Results indicate that 44.3% of the upazila falls under high to very high risk. Gabura, Padma Pukur, and Buri Goalini are identified as the most critically exposed unions. In contrast, northern unions including Bhurulia and Shyamnagar represent lower-risk zones, collectively covering 30.9% of the total upazila area. Field surveys in the highest-risk unions revealed a significant disease burden. Skin diseases (58%), scabies (36%), hypertension (30%), and gastric disorders (14%) were recorded among exposed populations, reflecting the compounded public health consequences of cyclone exposure. Based on a survey of 384 respondents in Shyamnagar Upazila, 46.7% of agricultural landowners have converted their land to aquaculture, primarily due to higher economic returns and reduced crop productivity. This figure rises to 68% in Gabura, coinciding with very high-risk zones. Limited sub-union-level spatial precision remains a methodological constraint, requiring further refinement in future research. The findings offer actionable spatial insights to guide cyclone shelter expansion, health infrastructure development, and climate-adaptive land-use planning in coastal Bangladesh.

Chapter One: Introduction

1.1 Background

Tropical cyclones rank among the most catastrophic natural hazards on Earth, posing existential threats to coastal communities through the convergence of destructive winds, torrential precipitation, and devastating storm surge inundation. Each year, approximately 90 tropical cyclones form across the world's ocean basins, collectively responsible for staggering human casualties and economic losses, disproportionately concentrated in low-lying coastal regions of the developing world (Bakkensen & Mendelsohn, 2019; Dube et al., 2009). The destructive capacity of these events extends far beyond the immediate period of cyclone landfall, generating cascading consequences, including infrastructure collapse, prolonged displacement, salinity intrusion, livelihood disruption, and deteriorating public health conditions, that can persist for years in affected communities. The convergence of climate change, accelerating sea-level rise, and rising sea surface temperatures are expected to substantially amplify the magnitude, frequency, and spatial reach of cyclone-induced coastal hazards in the coming decades (IPCC, 2007; Church et al., 2013). Storm surge flooding, widely regarded as the deadliest consequence of tropical cyclones, is anticipated to intensify significantly, threatening the lives, livelihoods, and long-term resilience of millions of people in vulnerable deltaic environments (Tessler et al., 2015). Prolonged exposure to post-cyclone conditions, including waterlogging, floodwater contamination, and elevated temperatures, further compounds health risks through increased incidence of waterborne diseases, respiratory illnesses, skin infections, and heat-related stress (Airi & Salgotra, 2026).

In this context, understanding the spatial dimensions of cyclone risk is no longer optional but an urgent scientific and policy imperative. Effective cyclone risk assessment demands an integrated framework that captures not only the physical intensity of the hazard but also the socio-economic conditions that determine community-level vulnerability and the institutional capacities that mediate mitigation. Composite index-based approaches combined with geospatial multi-criteria decision-making techniques, particularly the Analytical Hierarchy Process (AHP), have emerged as robust and widely applied tools for spatially explicit disaster risk assessment in data-constrained environments (Saaty, 2008; Malczewski, 2010). These approaches enable the systematic aggregation of heterogeneous physical, demographic, and infrastructural data into coherent spatial risk models that can directly inform targeted disaster risk reduction planning.

1.2 Coastal Area of Bangladesh, Cyclone Risk, and Storm Surge

Bangladesh stands as one of the most acutely vulnerable nations in the world to tropical cyclone impacts, a vulnerability rooted in the confluence of its low-lying deltaic topography, high population density, widespread poverty, and limited adaptive infrastructure. The country's distinctive funnel-shaped coastline along the Bay of Bengal acts as a natural amplifier of storm surge heights, dramatically intensifying the severity of coastal flooding during cyclonic events (Islam & Peterson, 2009). With a coastline stretching approximately 710 kilometres, Bangladesh's coastal zone is home to more than 35 million people whose livelihoods are intimately tied to climate-sensitive activities including agriculture, fisheries, and aquaculture. The historical record of cyclone impacts in Bangladesh is deeply sobering. The catastrophic 1991 cyclone claimed an estimated 138,000 lives and caused widespread destruction across the southeastern coast. More recently, Cyclone Sidr (2007) struck the southwestern coast with maximum sustained winds of 240 km/h, killing over 3,400 people and causing economic losses estimated at approximately USD 1.7 billion (Alam & Dominey-Howes, 2015). Cyclone Aila (2009) brought extensive embankment failures and prolonged inundation across Satkhira and Khulna districts, triggering land use transformation, particularly the forced conversion of agricultural land to shrimp aquaculture, and inflicting lasting socio-environmental damage on local communities (Sahana & Sajjad, 2019). Cyclone Amphan (2020), one of the most powerful Bay of Bengal cyclones on record, made landfall in the southwestern coastal region with sustained winds of 150 km/h, causing severe embankment breaches, crop losses, and widespread saline water intrusion across vulnerable coastal unions.

The southwestern coastal region, particularly Satkhira District, represents one of the most critically exposed zones in Bangladesh to cyclone-induced storm surges. This region is characterised by an intricate network of tidal rivers and estuarine channels connected to the Sundarbans mangrove ecosystem, low-lying polderised terrain, and communities whose agricultural livelihoods are severely constrained by recurring salinity intrusion and waterlogging. Rising sea levels in the Bay of Bengal further intensify these risks, progressively eroding the productive agricultural base and threatening the food security of millions of coastal inhabitants. The combination of physical exposure, socio-economic sensitivity, and institutional gaps in disaster preparedness makes the southwestern coast of Bangladesh a critical focal point for integrated cyclone risk assessment.

1.3 Problem Statement of the Research

Shyamnagar Upazila in Satkhira District, located adjacent to the Sundarbans mangrove forest and the Bay of Bengal, represents one of the most acutely vulnerable sub-district administrative units in coastal Bangladesh. The upazila is characterised by low-lying deltaic terrain, dense river networks, polder-based coastal protection systems, and a population almost entirely dependent on climate-sensitive livelihoods. Recurrent tropical cyclones — including Sidr (2007), Aila (2009), and Amphan (2020), have caused successive waves of embankment breaching, prolonged inundation, saline water intrusion, agricultural land degradation, and displacement, inflicting cumulative damage on the socio-economic fabric of local communities.

Despite this documented vulnerability, Shyamnagar Upazila suffers from a significant deficit of fine-scale, spatially explicit cyclone risk assessments that integrate physical hazard, socio-economic vulnerability, and mitigation capacity within a unified analytical framework. Existing assessments have predominantly been conducted at district or regional scales, obscuring critical intra-district and union-level variability in exposure, sensitivity, and adaptive capacity that is essential for targeting disaster risk reduction investments effectively. Furthermore, the linkage between spatial risk profiles and community-level health outcomes and livelihood transformation patterns, critical dimensions of cyclone impact, remains insufficiently examined. This research addresses this critical gap by developing an integrated, union-level cyclone risk assessment framework for Shyamnagar Upazila. The study constructs a Composite Vulnerability Index (CVI) integrating exposure, sensitivity, and resilience dimensions, develops AHP-weighted hazard and mitigation capacity indices incorporating physical and infrastructural variables including embankment failure, and generates a spatially explicit cyclone risk map validated against field observations. Additionally, the research examines the spatial correspondence between risk classification and community-level health impacts and agricultural land use transformation, providing a comprehensive empirical basis for evidence-based coastal risk management and disaster preparedness planning.

1.4 Research Gap

A review of the existing literature reveals several important gaps that this study is designed to address. First, while geospatial multi-criteria approaches to cyclone risk assessment have been applied in Bangladesh (Hoque et al., 2018, 2019; Sahana & Sajjad, 2019), most studies have been conducted at coarse spatial scales, district or regional level, that are insufficient for capturing the fine-scale heterogeneity in vulnerability and adaptive capacity among individual unions. This scale limitation significantly constrains the applicability of research findings for targeted community-level disaster risk reduction planning.

Existing integrated risk frameworks have often treated vulnerability and hazard components separately, or have excluded critical physical variables such as embankment failure, which plays a decisive role in determining flood extent and duration in polder-protected coastal regions of Bangladesh (Mallick et al., 2011; Roy & Blaschke, 2013). The systematic exclusion of embankment integrity as a hazard determinant represents a significant methodological gap in prior assessments.

Despite growing recognition that cyclone impacts generate cascading socio-environmental consequences, including long-term salinity intrusion, livelihood transformation, rural poverty dynamics, and public health deterioration, integrated risk frameworks that simultaneously examine these cascading dimensions alongside physical risk mapping remain scarce (Sebesvari et al., 2016; Hazra et al., 2017). The spatial relationship between risk classification and community-level health burdens and agricultural land use change has received particularly limited attention in the existing literature.

Most prior vulnerability studies have relied on secondary administrative data without field-based validation, limiting confidence in their empirical accuracy and relevance to local conditions. This study addresses all four gaps through a comprehensive, field-validated, union-level assessment that integrates physical hazard, composite vulnerability, health outcomes, and land use transformation within a unified geospatial framework.

1.5 Research Questions

This study is guided by the following central research questions:

1. What is the spatial distribution of storm surge flood inundation risk across different unions of Shyamnagar Upazila under varying water level scenarios?
2. How do exposure, sensitivity, and resilience dimensions of vulnerability vary spatially at the union level, and what is the resulting Composite Vulnerability Index distribution?
3. What is the integrated tropical cyclone risk profile of Shyamnagar Upazila when hazard intensity, composite vulnerability, and mitigation capacity are jointly assessed using an AHP-based geospatial framework?
4. To what extent do spatial cyclone risk levels correspond with community-level health impact patterns and agricultural land use transformation trends?
5. What targeted adaptation and disaster risk reduction strategies are most urgently required in high-risk unions of Shyamnagar Upazila?

1.6 Research Objectives

1. Develop a cyclone risk map using GIS and AHP
2. Integrate hazard, vulnerability, and mitigation capacity
3. Identify high-risk zones and exposed areas
4. Assess health impacts in vulnerable communities
5. Analyze agricultural land-use change in relation to risk

1.7 Rationale of the Study

The rationale for this research emerges from the critical intersection of escalating physical risk, acute socio-economic vulnerability, and persistent knowledge gaps in cyclone risk assessment for Shyamnagar Upazila. Bangladesh's southwestern coastal region is among the most systematically exposed and socio-economically vulnerable areas on the planet to cyclone-induced hazards, yet fine-scale risk intelligence necessary for evidence-based disaster management remains severely limited. This knowledge deficit has direct and measurable consequences: disaster risk reduction investments are allocated without the spatial precision needed to prioritise the most vulnerable communities, emergency preparedness systems are not calibrated to local risk gradients, and climate adaptation planning proceeds without a rigorous empirical foundation.

By developing a comprehensive, multi-dimensional, union-level cyclone risk framework validated through primary field data, this study generates both scientific knowledge and actionable intelligence for policymakers, disaster management authorities, and development practitioners. The integration of health and livelihood dimensions within the risk assessment framework ensures that findings capture the full human cost of cyclone exposure rather than limiting analysis to physical hazard metrics alone. This holistic approach aligns with the Sendai Framework for Disaster Risk Reduction 2015–2030, which explicitly calls for multi-hazard, people-centred risk assessments that support targeted risk reduction investments and build community resilience.

1.8 Scope of the Research

This study is geographically bounded to Shyamnagar Upazila in Satkhira District, southwestern coastal Bangladesh, encompassing 12 unions as the primary unit of spatial analysis. The research focuses on tropical cyclone risk with particular emphasis on storm surge flooding as the dominant hazard mechanism. Temporally, the hazard assessment draws on historical cyclone track data from 2000 to 2026 and precipitation records from 2000 to 2022, while the land use analysis is based on Sentinel-2 imagery from 2024 and field surveys conducted between October and December 2025.

The scope of the vulnerability assessment encompasses physical exposure indicators (flood inundation risk, slope, drainage proximity and density, storm surge frequency), socio-economic sensitivity indicators (population density, agricultural land, built-up density, child population, water access, poverty), and resilience indicators (literacy rate, housing quality, communication

infrastructure). Hazard assessment incorporates Cyclone Amphan wind speed data, historical cyclone frequency, precipitation intensity, Composite Vulnerability Index scores, and embankment failure records. Mitigation capacity is assessed through cyclone shelter accessibility, health facility proximity, vegetation cover (NDVI), and warning system effectiveness.

The research does not extend to dynamic hydrodynamic modelling of flood velocities or probabilistic cyclone track forecasting; these represent directions for future research. Similarly, household-level socio-economic surveys were not conducted within the scope of this study, with union-level administrative data used as the primary socio-economic data source.

1.9 Rationale for Undertaking this Research Work

The decision to focus specifically on Shyamnagar Upazila is motivated by several converging considerations. Shyamnagar represents one of the most systematically and repeatedly impacted sub-districts in coastal Bangladesh, having experienced major cyclone events including Sidr, Aila, and Amphan within the past two decades. Its proximity to the Sundarbans, its dependence on agriculture and fisheries livelihoods, and its polder-based coastal protection infrastructure, which is highly vulnerable to embankment failure, create a distinctive and compelling case study for integrated risk assessment. Furthermore, the union-level administrative scale selected for this study provides a spatial resolution that is directly actionable for local government bodies, upazila disaster management committees, and national disaster risk reduction authorities. Unlike district-level assessments, union-level risk maps can directly inform decisions about cyclone shelter placement, embankment reinforcement priorities, health infrastructure investment, and community-level early warning system design. The choice of this scale therefore reflects a deliberate commitment to producing research that is not only scientifically rigorous but also operationally meaningful for coastal risk governance in Bangladesh.

Finally, this research is motivated by the recognition that the cascading relationship between physical cyclone hazard, socio-economic vulnerability, public health burden, and livelihood transformation in coastal Bangladesh is deeply underexplored at fine spatial scales. By establishing this empirical linkage through an integrated, field-validated framework, the study makes a distinctive scientific contribution that advances both the theoretical understanding of coastal risk dynamics and the practical tools available to disaster managers and policymakers in one of the world's most climate-exposed regions.

Chapter Two: Literature Review

2.1 Climate Change, Cyclone, and Storm Surge

The relationship between anthropogenic climate change and tropical cyclone behaviour has become one of the most intensively studied and policy-relevant topics in contemporary climate science. There is now robust scientific consensus that rising sea surface temperatures, a direct consequence of greenhouse gas-driven climate warming, are intensifying the destructive potential of tropical cyclones, increasing the likelihood of rapid intensification events and expanding the latitudinal range within which intense cyclones can form (IPCC, 2007; Church et al., 2013). Projections from multiple coupled climate models indicate that while the total number of tropical cyclones globally may not increase substantially, the proportion of storms reaching the highest intensity categories is expected to rise significantly, with potentially devastating consequences for coastal populations (Tessler et al., 2015).

Storm surge, the rapid coastal flooding generated by the combination of cyclone-driven wind stress and low atmospheric pressure, represents the most lethal and economically destructive consequence of tropical cyclone landfall in low-lying coastal regions. The height and inland reach of storm surges are determined by a complex interplay of cyclone intensity, translation speed, angle of approach, coastal geometry, and nearshore bathymetry (Islam & Peterson, 2009). Accelerating sea-level rise associated with climate change is expected to substantially amplify storm surge heights and inland penetration in the coming decades, progressively inundating areas that are currently above the threshold of cyclone-induced flooding. In deltaic environments such as the Bengal delta, this risk is compounded by ongoing land subsidence driven by sediment compaction, groundwater extraction, and reduced sediment supply resulting from upstream dams (Tessler et al., 2015). Together, these interacting processes are expected to significantly expand the population exposed to catastrophic storm surge flooding under plausible mid-century climate scenarios.

Prolonged exposure to elevated temperatures, flooding, and water contamination in the aftermath of cyclone events further increases health risks across affected communities, including heat stress, respiratory illnesses, and waterborne diseases (Airi & Salgotra, 2026). The public health dimension of cyclone impacts is therefore inseparable from physical hazard assessment, underscoring the need for integrated risk frameworks that capture both environmental and human dimensions of disaster risk.

2.2 Understanding Cyclones and Risk

Tropical cyclones are intense atmospheric vortices that derive their energy from the latent heat released during the condensation of moisture evaporated from warm ocean surfaces. Their formation requires a threshold sea surface temperature typically above 26–27°C, sufficient atmospheric instability, and low vertical wind shear (Emanuel, 2005). Climate change is modifying each of these environmental preconditions in ways that are expected to alter both the frequency distribution and intensity spectrum of tropical cyclones in major ocean basins, including the Bay of Bengal.

The Bay of Bengal is among the world's most cyclone-active ocean basins, generating approximately 5–6% of global tropical cyclones annually but accounting for a disproportionately large fraction of global cyclone-related mortality due to the extreme vulnerability of the densely populated, low-lying coastlines surrounding it (Dube et al., 2009). Warming sea surface temperatures in the northern Bay of Bengal are creating increasingly favourable thermodynamic conditions for cyclone formation and intensification, particularly during the pre-monsoon (April–May) and post-monsoon (October–December) cyclone seasons. Recent decades have witnessed a notable trend toward more intense and rapidly intensifying cyclones in the Bay of Bengal, a pattern consistent with theoretical expectations from climate change science (Bakkensen & Mendelsohn, 2019).

From a vulnerability perspective, the interaction between intensifying physical hazard and socio-economic exposure is particularly alarming in coastal Bangladesh. The country's low-lying deltaic geography, with large areas less than 1 metre above mean sea level, means that even marginal increases in storm surge height can translate into catastrophic expansions of inundation extent. This physical sensitivity is compounded by high population densities, widespread poverty, and infrastructure deficits that limit community adaptive capacity and post-disaster recovery. Understanding the mechanistic relationships between climate change, cyclone behaviour, and coastal vulnerability is therefore foundational to developing realistic projections of future risk and designing adaptive interventions appropriate to the scale and urgency of the challenge.

2.3 Impact of Cyclones on Health and Livelihood

The impacts of tropical cyclones on human health and livelihoods in coastal Bangladesh extend far beyond the immediate mortality and physical destruction of the storm event itself. Post-cyclone environments generate a cascade of secondary health hazards that can persist for months or years following landfall. Contamination of surface water and groundwater sources by saline intrusion, floodwater, and displaced sanitation infrastructure creates conditions highly conducive to waterborne diseases including cholera, typhoid, and gastroenteritis (Sahana & Sajjad, 2019). Prolonged exposure to stagnant floodwater, combined with disrupted healthcare access, elevated rates of skin infections, scabies, and other dermatological conditions among storm-affected populations. Mental health impacts, including anxiety, depression, and post-traumatic stress, are increasingly recognized as a significant but historically underreported dimension of the cyclone health burden.

The livelihood impacts of cyclones in coastal Bangladesh are equally severe and often long-lasting. Agriculture, the dominant livelihood system for rural coastal households, is particularly vulnerable to cyclone impacts through crop destruction, embankment breaching and subsequent saline inundation of agricultural land, and long-term soil salinization that reduces land productivity for seasons or years after the cyclone event (Sebesvari et al., 2016). In Shyamnagar and adjacent areas, the aftermath of Cyclone Aila triggered the forced conversion of extensive areas of agricultural land to shrimp aquaculture, as salinity-damaged soils became unsuitable for traditional rice cultivation but economically viable for brackish water shrimp farming (Sahana & Sajjad, 2019). While shrimp aquaculture has provided some households with an income source, it has simultaneously displaced agricultural laborers, increased income inequality, and created ecologically fragile monoculture production systems highly sensitive to market fluctuations and disease outbreaks.

Fisheries, another critical livelihood system in the coastal zone, are disrupted during cyclone events by destruction of fishing vessels, gear, and landing infrastructure, with recovery periods often extending for several months. The cumulative effect of repeated cyclone impacts, combined with chronic stressors including salinity intrusion, waterlogging, and coastal erosion, is a progressive erosion of livelihood assets, deepening poverty, and increasing out-migration from the most exposed coastal unions.

2.4 Cyclone Risk in Shyamnagar

Shyamnagar Upazila in Satkhira District occupies a frontline position in Bangladesh's cyclone risk landscape, characterized by its direct exposure to Bay of Bengal storms, its location adjacent to the Sundarbans mangrove ecosystem, and its dependence on climate-sensitive livelihoods that are acutely sensitive to cyclone impacts. The upazila's geography, low-lying deltaic terrain intersected by a dense network of tidal rivers and protected by an extensive polder system, creates a distinctive risk profile in which embankment integrity is a critical determinant of flood extent and duration during cyclone events.

Historical cyclone records demonstrate that Shyamnagar has experienced repeated major cyclone impacts at intervals of approximately 5–10 years over the past several decades. Cyclone Sidr (2007), Cyclone Aila (2009), and Cyclone Amphan (2020) each caused extensive embankment breaches, inundation of agricultural land, damage to settlements, and significant human suffering in the upazila. The cumulative impact of these events has progressively degraded the coastal protection infrastructure, reduced agricultural productivity through salinity intrusion, and eroded household-level livelihood resilience across the most exposed unions (Alam & Dominey-Howes, 2015; Sahana & Sajjad, 2019). Post-cyclone conditions in Shyamnagar are particularly protracted due to the slow drainage of floodwaters in the low-lying polder system and the persistence of saline water in formerly productive agricultural land.

Dasgupta et al. (2014) identified the southwestern coastal districts of Bangladesh, including Satkhira, as among the areas most likely to experience increasing cyclone-related losses under future climate scenarios, driven by the combination of sea-level rise, intensifying storm surge, and high socio-economic vulnerability. Field-based research in Shyamnagar has further documented the complex interactions between physical hazard, land use change, and livelihood transformation that make the upazila a critical hotspot for investigating cascading cyclone risk dynamics in coastal deltaic environments (Auerbach et al., 2015; Rahman et al., 2019).

2.5 Long-Term Impacts of Cyclones

The long-term consequences of major cyclone events in coastal Bangladesh extend well beyond the immediate disaster period, generating enduring transformations in environmental conditions, land use systems, social structures, and economic trajectories that can persist for decades. Understanding these long-term dynamics is essential for designing resilience-building strategies that address not only short-term disaster response but also the structural conditions of vulnerability that accumulate between cyclone events. One of the most consequential long-term impacts of cyclones in the southwestern coastal zone is progressive soil salinization. When embankments are breached during storm surge events, saline water inundates formerly fresh or brackish agricultural land, depositing salt that penetrates the soil profile and impairs plant growth for subsequent growing seasons. In areas of Shyamnagar and adjacent upazilas affected by Cyclone Aila (2009), agricultural land remained waterlogged and saline for up to three years following the cyclone event, forcing prolonged displacement of farming households and triggering permanent land use transitions in the worst-affected areas (Sahana & Sajjad, 2019). This salinity legacy effect represents a form of chronic environmental damage that is qualitatively different from, and in many ways more socio-economically damaging than, the acute destruction of the cyclone itself. Long-term cyclone impacts also operate through their effects on migration and demographic change. Repeated cyclone events that destroy agricultural livelihoods and housing drive both temporary displacement and permanent out-migration from the most exposed coastal communities, progressively altering the demographic composition of cyclone-affected areas (Hazra et al., 2017). These demographic shifts compound the physical vulnerabilities created by cyclone damage, creating feedback loops that progressively undermine community resilience.

Ecosystem degradation represents another critical long-term consequence of repeated cyclone impacts. The Sundarbans mangrove forest, which provides natural storm protection for Shyamnagar and adjacent coastal areas, has experienced repeated damage from major cyclones including Sidr and Amphan, with recovery of cyclone-damaged mangrove stands requiring several years under favourable conditions (Das & Vincent, 2009). Progressive degradation of the Sundarbans ecosystem reduces the natural storm attenuation function that this forest provides, creating positive feedback between cyclone damage, ecosystem deterioration, and increasing coastal vulnerability that underscores the importance of ecosystem-based adaptation as a component of coastal risk management.

Chapter Three: Methodology

3.1 Methods

Methodological flowchart of the cyclone risk assessment are in (Fig. 1):

Methodological flowchart of the cyclone risk assessment

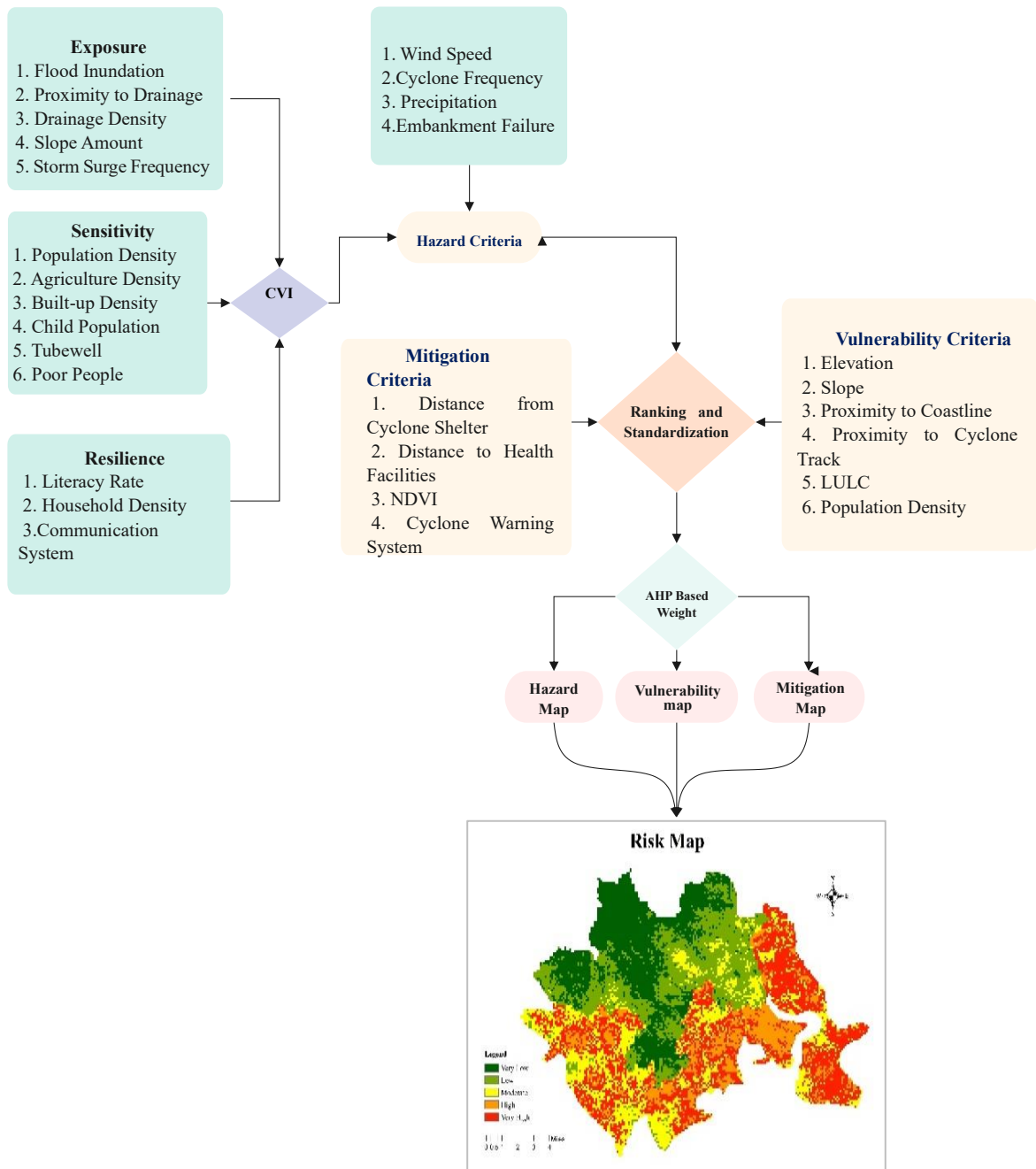


Figure 1 Methodological flowchart of the cyclone risk assessment approach

The AHP has a strong capability to efficiently incorporate and aggregate multi-criteria and present findings effectively (Malczewski, 1999; Mani Murali et al., 2013). Few risk assessment equations are found in the literature for mapping the risk of natural hazards (Dewan, 2013a; Eckert et al., 2012; Hoque et al., 2017; Masood and Takeuchi, 2012). Selecting an advanced and complete risk equation with essential components is essential for good and accurate risk mapping. In this study, Eq. 1 was adopted to map tropical cyclone risks following an extensive review of the current literature (Dewan, 2013a; Hoque et al., 2018a; Rashid, 2013).

$$Risk = Hazard \times Vulnerability \times exposure \times mitigation / capacity \dots\dots\dots 1$$

3.2 Study Area

Shyamnagar Upazila, located in the south-western coastal region of Bangladesh within Satkhira District, extends approximately between 21°36'N to 22°24'N latitude and 89°00'E to 89°19'E longitude, and lies adjacent to the Sundarbans and the Bay of Bengal (Fig. 2). This strategic coastal position makes the area highly exposed to tropical cyclone hazards and associated storm surge flooding. The upazila is characterized by low-lying deltaic terrain, dense river networks, and polder-based coastal protection systems, which significantly influence its hydro-geomorphological vulnerability. Shyamnagar was selected for this study due to its high susceptibility to recurrent cyclones, embankment breaching, prolonged waterlogging, and severe salinity intrusion, all of which critically affect agricultural productivity, freshwater availability, and ecosystem stability. Historically, extreme events such as Cyclone Sidr, Cyclone Aila, and Cyclone Amphan have caused extensive inundation, infrastructure damage, displacement, and long-term socio-economic disruption in this region. The dominance of climate-sensitive livelihoods, particularly agriculture and shrimp aquaculture, further exacerbates the vulnerability of local communities. Additionally, post-cyclone conditions often trigger significant public health risks, including waterborne and skin diseases, due to poor sanitation and stagnant water conditions. These combined physical, environmental, and socio-economic stressors make Shyamnagar a critical hotspot for assessing cyclone-induced vulnerability, exposure, and adaptive capacity in coastal Bangladesh (Dasgupta et al., 2014; Auerbach et al., 2015; Rahman et al., 2019; Islam & Hasan, 2016).

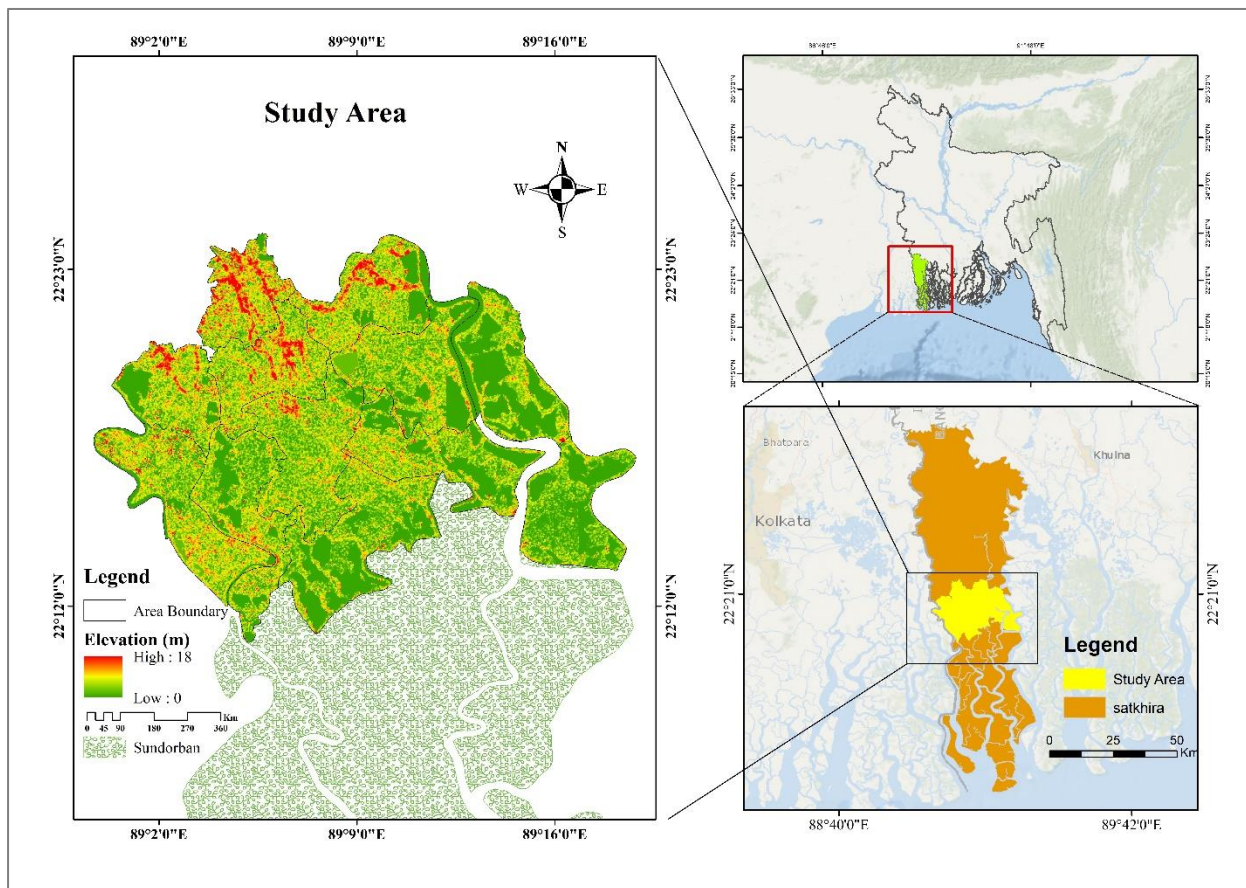


Figure 2: Study Area

3.3. Flood Inundation

Flood inundation mapping was carried out using Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM) with 30 m spatial resolution. The DEM was processed in ArcGIS (ArcMap 10.8) to simulate potential storm surge inundation under multiple water level scenarios. Initially, the DEM was projected to a uniform coordinate system and pre-processed using the Fill tool to remove sinks and ensure hydrological consistency. Flood inundation scenarios were generated by applying elevation-based thresholding corresponding to different storm surge water levels (0.5 m, 1 m, 2 m, 3 m, 4 m, and 5 m). For each scenario, raster reclassification was performed where cells with elevation less than or equal to the threshold were classified as inundated (value = 1), and those above the threshold were classified as non-inundated (value = 0). Following Sahana and Sajjad (2019), the Flood Inundation Risk (FIR) was estimated by integrating multi-scenario inundation extents using a composite index approach. In this study, a modified aggregation method

was applied using equal-weight averaging of different inundation levels to represent overall flood susceptibility:

$$FIR = \sqrt{\frac{a+b+c+d+e+f}{6}} \dots\dots\dots 2$$

where a–f represents the percentage of inundated area under 0.5 m, 1 m, 2 m, 3 m, 4 m, and 5 m storm surge scenarios, respectively. This approach allows the representation of cumulative flood exposure across multiple storm surge intensities, where higher FIR values indicate higher probability and persistence of inundation under different flood conditions. The resulting FIR values were then classified into five flood hazard categories (very low, low, moderate, high, and very high) using percentile-based classification thresholds. The index, then, was used as an important indicator for assessing storm surge exposure index.

3.4. Vulnerability Indicators and Construction of the Vulnerability Index

Based on available data and our concepts of vulnerability, we have selected the following indicators for the CVI (Table 1):

- **Exposure indicators:** Flood inundation risk, proximity to drainage, drainage density, slope, and average storm surge frequency.
- **Sensitivity indicators:** population density, agricultural land cover, built-up density, child population proportion, access to tube well or borehole water, and poor people.
- **Resilience indicators:** literacy rate, Communication System.

Proximity to drainage significantly influences the spread and magnitude of coastal flooding (Fernández and Lutz, 2010). Flood Inundation map which area is risk during Cyclone time storm surge (Fig. 3a). Buffer zones along rivers also affect ground slope and overland flow, making them directly relevant to human settlement in riparian areas. A drainage proximity map was generated using 5km interval buffers, divided into five zones. Generally, greater distance from drainage lines corresponds to lower flood exposure (Fig.3b). Drainage density the ratio of total drainage length to cell area, is a key variable contributing to flood occurrence. The drainage network was digitized from topographic sheets (1:50,000) and verified using Google Earth imagery, then converted to a

drainage density layer in ArcGIS. Five density zones were established (below 2 km/km², 2-5, 5-10, 10-20, and above 20 km/km²) to assess their influence on coastal flooding (Fig.3c).

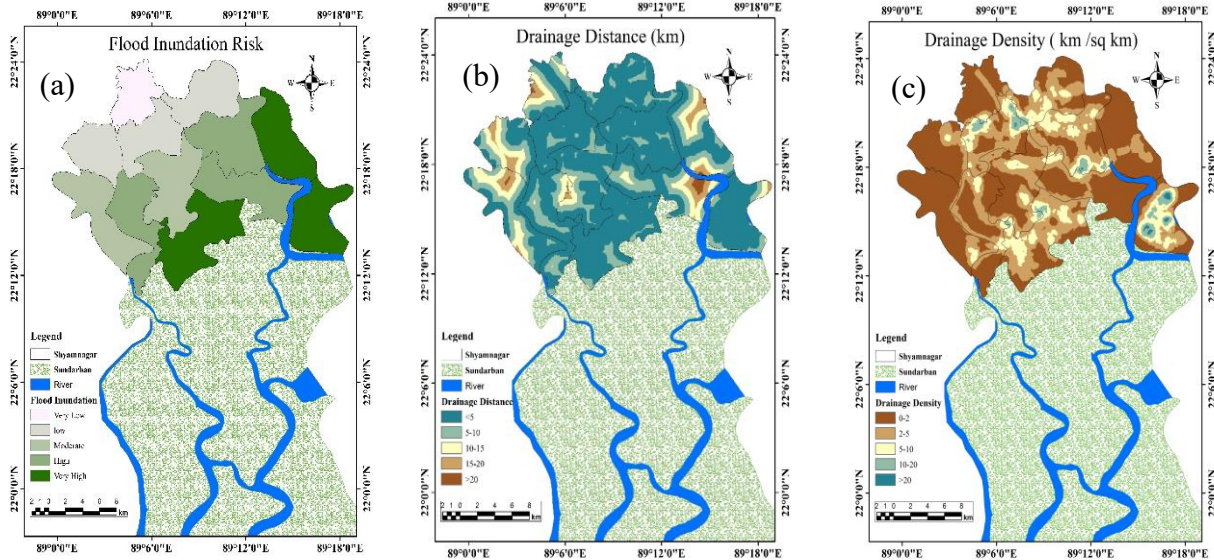


Figure 3. (a) Flood Inundation (b) Proximity to Drainage (c) Drainage Density

Slope governs storm water velocity and the degree of flood exposure. Gentle slopes are particularly susceptible to tsunamis and cyclonic storm surges (Klein et al., 2001; Hegde and Reju, 2007). Slope values were derived from a Digital Elevation Model (DEM), with zero-value surface errors corrected using ArcGIS's fill function. The region was divided into five slope categories at 1-metre intervals from mean sea level. Slope of the study area was derived from NASADEM (1 arc-second resolution) provided by NASA JPL (2021). The DEM was pre-processed in ArcGIS by applying the Fill tool to correct spurious depressions and eliminate zero-value inconsistencies (Fig.4a). Subsequently, slope was generated in degrees using the *Slope tool* in the Spatial Analyst extension. The resulting slope values were classified into five categories based on natural breaks and terrain characteristics of the study area, namely: 0–0.83°, 0.84–2°, 2.1–3.5°, 3.6–7.3°, and 7.4–42° (Fig.4b).

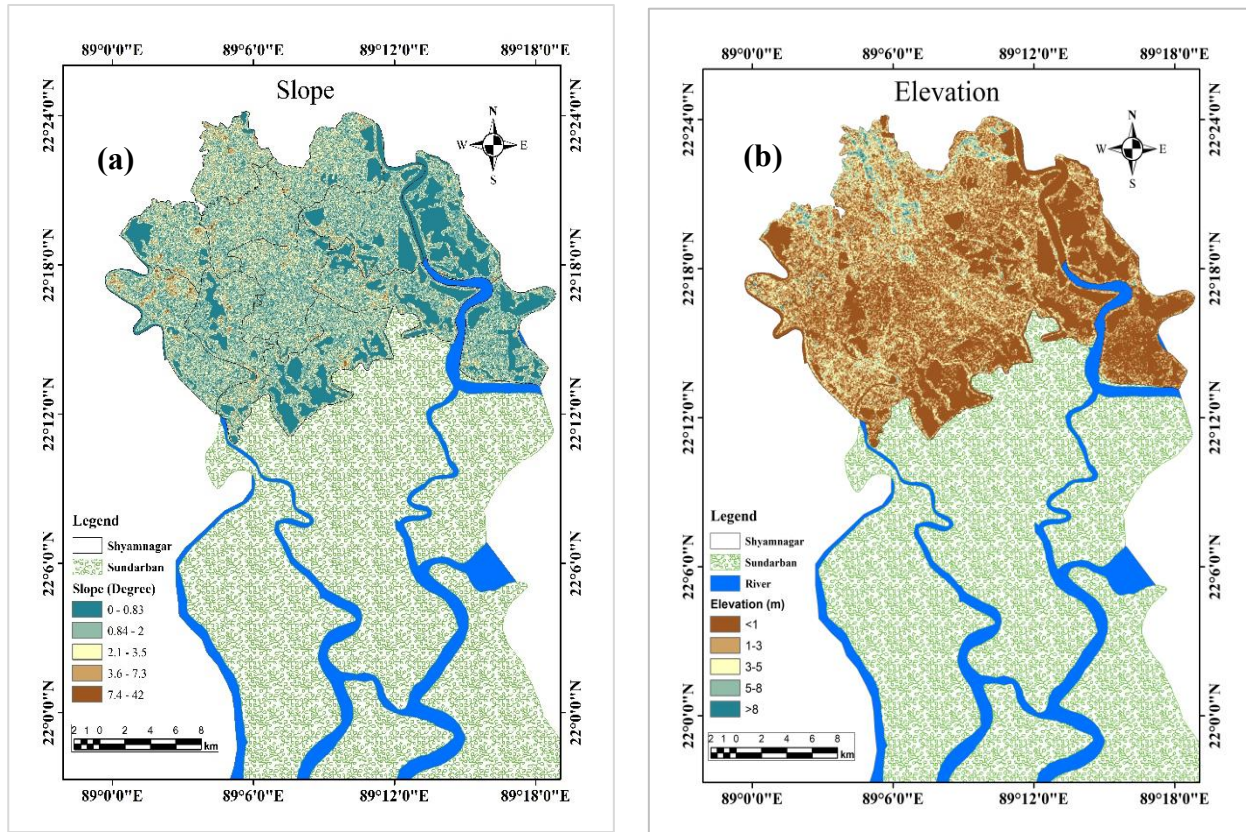


Figure 4. (a) Slope (b) Elevation

. Storm surge frequency, an important indicator of coastal exposure, was assessed based on historical information collected through field surveys and consultation with local authorities. The frequency of cyclone-induced storm surge events was categorized into five classes (<1, 2, 3, 4, and >5 events), (Fig. 5) representing increasing levels of exposure. Areas experiencing higher frequencies of storm surge events were considered more vulnerable due to repeated hazard occurrence. Each exposure variable was classified into five intensity classes and ranked on a scale of 1 to 5 (Table 1) . The Exposure Index for each village was calculated as:

$$EI = \frac{(E_1 + E_2 + E_3 + E_4 + E_5)}{5} \dots\dots\dots 3$$

where E_1 = flood inundation risk, E_2 = proximity to drainage, E_3 = drainage density, E_4 = slope, and E_5 = average storm surge height.

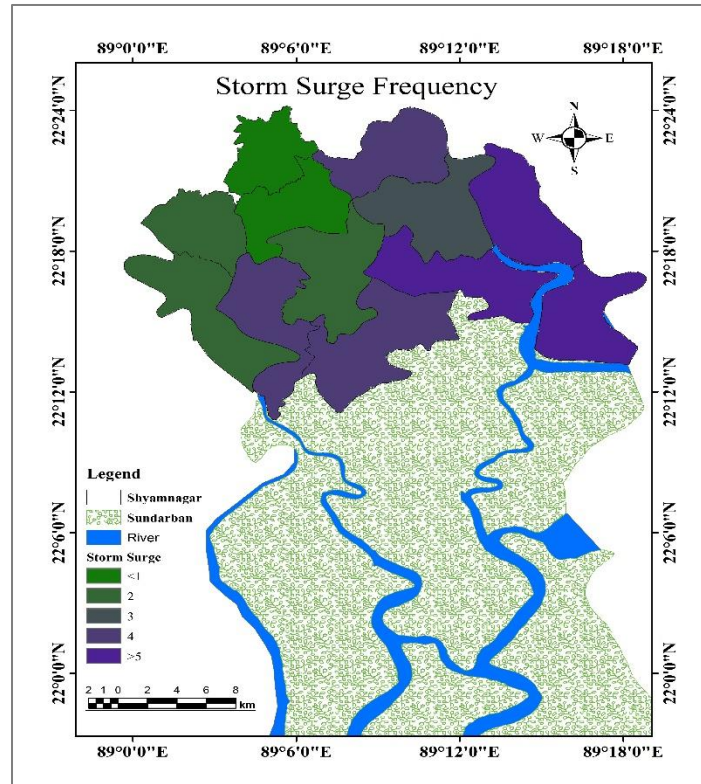


Figure 5: Storm Surge Frequency

Table. 1 Components of composite vulnerability index and their statistics

Components	Indicators	Source	Class	Rank
Exposure	Level of flood inundation	NASA JPL (2021). <i>NASADEM Merged DEM Global 1 arc second V001</i> . Distributed by OpenTopography. https://doi.org/10.5069/G93T9FD9 . Accessed 2026-04-16	Very Low	1
			Low	2
			Moderate	3
			High	4
			Very High	5
	Proximity to the drainage	Global Map Initiative , Local Government Engineering Department (LGDE)	> 5 km	1
			5-10 km	2
			10-15 km	3
			15-20 km	4
			>20 km	5
	Drainage density	Global Map Initiative , Local Government Engineering Department (LGDE)	< 2 km/sq km	1
			2-5	2
			5-10	3
			10-20	4
			>20	5
	Slope amount	NASA JPL (2021). <i>NASADEM Merged DEM Global 1 arc second V001</i> . Distributed by OpenTopography. https://doi.org/10.5069/G93T9FD9 . Accessed 2026-04-16	0-0.83	1
			0.84-2	2
			2.1-3.5	3
			3.6-7.3	4
			7.4-42	5
	Storm Surge Frequency	Field Survey and Local Authorities	<1	1
			2	2
			3	3
			4	4

			> 5	5
Sensitivity	Population density	Union profilr.gov.bd	< 600	1
			670-775	2
			775-969	3
			970-1044	4
			>1044	5
	Agricultural land (%)	Sentinel-2 imagery (ESA), 2024	Very Low	1
			Low	2
			Moderate	3
			High	4
			Very High	5
	Built-up density	Sentinel-2 imagery (ESA), 2024	-0.78- -0.39	1
			-0.38- -0.24	2
			-0.23 – -0.1	3
			-0.09– 0.034	4
			0.035-0.61	5
	Child population (%)	City Population (Shyamnagar, Khulna)	< 5531	1
			5532–6679	2
			6680–7956	3
			7957 – 9299	4
			> 9300	5
	Tubewell/Borehole water facility (%)	Union profilr.gov.bd	> 421	1
			181 – 420	2
			43–180	3
			6–42	4
			< 5	5
	Poor People (%)	Union profilr.gov.bd	< 800	1
			1277-4022	2
			4023–6555	3
			6556–11855	4
			> 11856	5
Resilience	Literacy rate	Union profilr.gov.bd	37-40	1
			41-45	2
			46-60	3
			61-70	4
			71-77	5
	Households pakka (%)	Union profilr.gov.bd	< 480	1
			480-580	2
			580-680	3
			680-780	4
			> 780	5
	Communication System	Field Survey and Local Authorities	Very poor	1
			poor	2
			Moderate	3
			Good	4
			Excellent	5

Sensitivity denotes those elements that influence the intensity of hazard risk. Population density, settlement characteristics, and agricultural land significantly increase the severity of coastal flood hazards (Feng et al., 2013). In this study, sensitivity was assessed using six key indicators: population density, agricultural land, built-up density, child population, tubewell/borehole water facility, and poor population. Population density, tubewell facility, and poor population data were collected from Union Profile (gov.bd) (Fig.6),

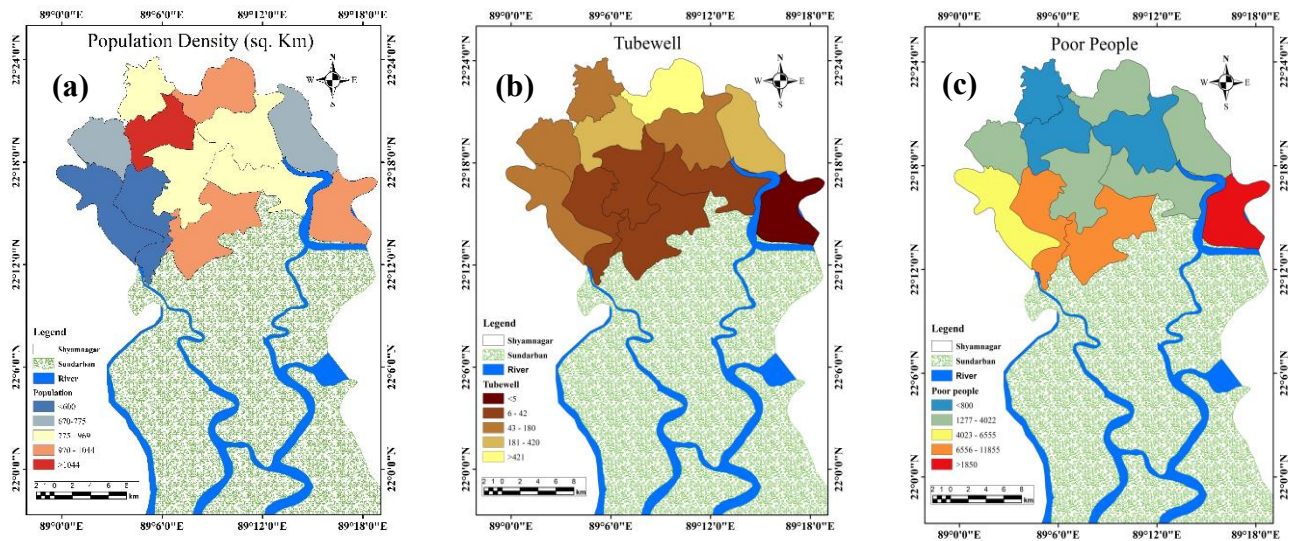


Figure 6: (a) Population density, (b) Tubewell facility, and (c) Poor population

A agricultural land and built-up density were derived from Sentinel-2 imagery (2024) (Fig.7).

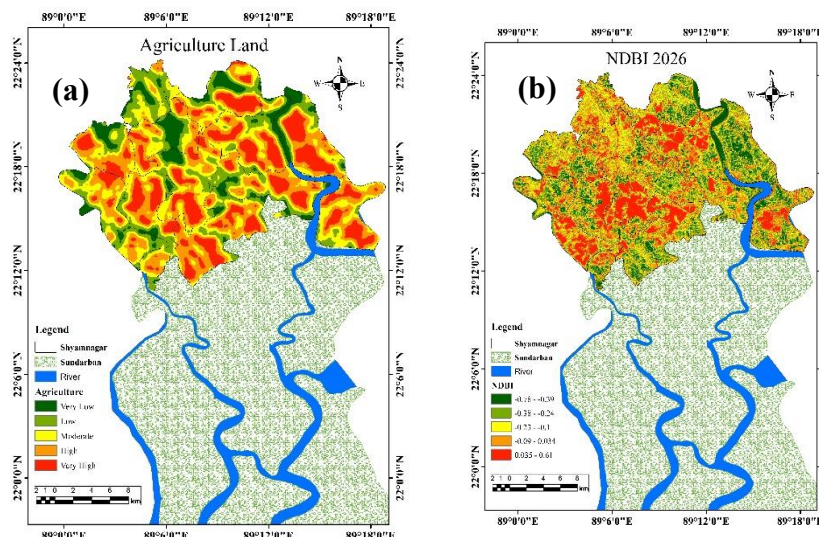


Figure 7: (a) Agricultural Land (b) NDBI 2026

Child population data were obtained from City Population (based on Bangladesh Bureau of Statistics) (Fig.8).

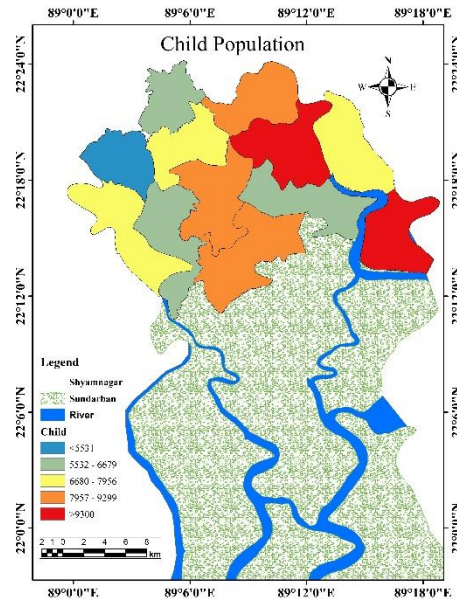


Figure 8: Child Population Map

Higher population density, greater extent of agricultural land, and increased built-up density contribute to higher exposure and damage potential. Children are considered more vulnerable to hazards; therefore, a higher proportion of child population indicates increased sensitivity. Similarly, limited access to safe drinking water (tubewell/borehole facility) and a higher proportion of poor population increase susceptibility to cyclone impacts. Each variable was classified into five categories representing increasing levels of sensitivity and reclassified into a standardized scale from 1 (very low) to 5 (very high). The Sensitivity Index (SI) was calculated using an equal-weight approach:

$$SI = \frac{S_1 + S_2 + S_3 + S_4 + S_5 + S_6}{6} \dots \dots \dots 4$$

where S_1 to S_6 represent the ranked values of population density, agricultural land, built-up density, child population, tubewell facility, and poor population, respectively.

Resilience is defined as the capacity of a system to anticipate, withstand, and recover from hazard impacts through socio-economic and infrastructural support systems (Balica et al., 2012; Aydın and Uysal, 2014). In this study, resilience was assessed using three key indicators: literacy rate, households with pucca housing, and communication system. Literacy rate data were obtained from Union Profile (gov.bd), while household information on pucca housing was also collected from the same source. Communication system quality was determined through field surveys and consultation with local authorities. Higher literacy rates, improved housing conditions, and efficient communication systems indicate stronger adaptive and coping capacity against cyclone-induced hazards. Each indicator was classified into five categories ranging from very low (1) to very high (5) based on their relative contribution to resilience (Fig.9).

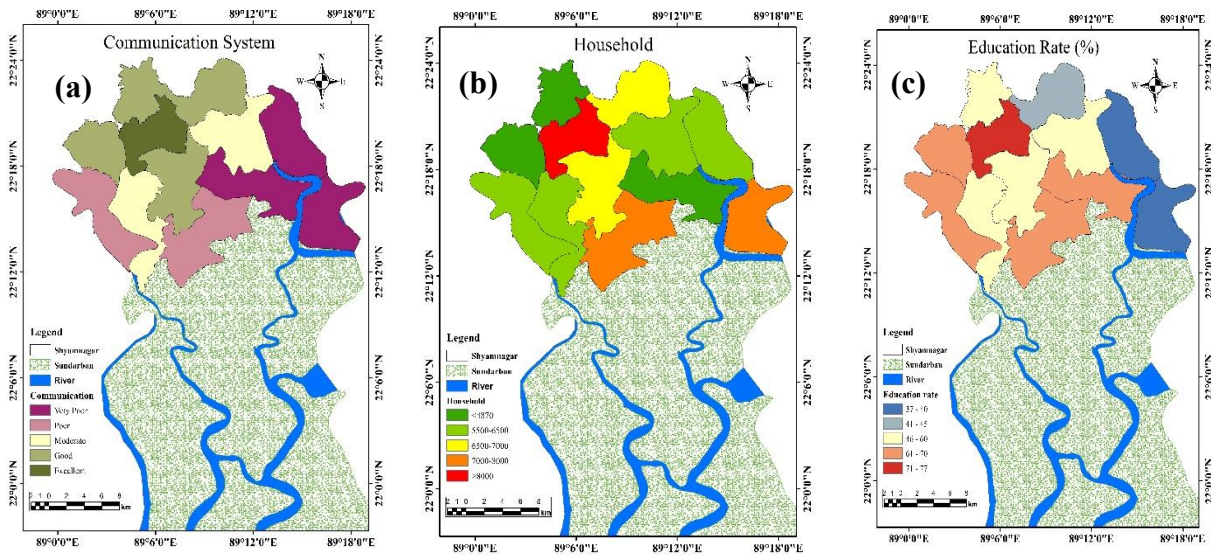


Figure 9: (a) Communication System (b) Household (c) Education Rate

The Resilience Index (RI) was calculated using an equal-weight approach:

$$RI = \frac{R_1 + R_2 + R_3}{3} \dots\dots\dots 5$$

where R_1 , R_2 , and R_3 represent the ranked values of literacy rate, pucca households, and communication system, respectively.

Finally, composite vulnerability index using its three component indices (exposure index, sensitivity index and resilience index) was determined as:

$$CVI = Exposure + Sensitivity - Resilience \dots\dots\dots 6$$

3.5. Risk Evaluation Criteria, Alternatives And Mapping

In this study, criteria were selected following an intensive literature review as well as by considering data accessibility and their relevance to tropical cyclone risks. The development of spatial layers for each criterion was performed using GIS and remote sensing techniques. We generated 15 spatial criteria layers for the risk components of vulnerability, hazard and mitigation capacity. A 10 m × 10 m cell size was determined as a spatial resolution of each raster layer. Remote sensing ArcGIS (version 10.8) were used to process and prepare all the criteria layers. The natural break statistical methods were used to classify the produced maps. This statistical method was found more consistent and effective to exhibit the spatial pattern of risk in the study area (Baeza et al., 2016; Tehrany et al., 2014). The characteristics and mapping approaches for the selected criteria are detailed in the succeeding subsections

3.5.1. Criteria for vulnerability mapping

Several criteria influence the vulnerability and exposure of any area to tropical cyclone impacts (Hoque et al., 2018a). In the present study, six criteria related to tropical cyclone vulnerability and exposure were chosen. Mainly focused on the physical aspect, the vulnerability criteria included elevation, slope, proximity to coast and cyclone track. Exposure criteria comprised land use and land cover as well as population density.

Elevation and slope play a crucial role in tropical cyclone vulnerability assessment (Dewan, 2013b; Li and Li, 2013). Areas with the characteristics of low elevation and slope are considered highly vulnerable to tropical cyclones, whereas areas with high slope and elevation are considered less vulnerable (Rao et al., 2013). In this study, a 10-m spatial resolution digital elevation model (DEM) was produced from the topographic sheets (scale 1:25000) to generate the elevation and slope map (Fig. 10 a, b). This DEM was obtained from the NASA JPL (2021). NASADEM Merged DEM Global 1 arc second V001. Distributed by OpenTopography.

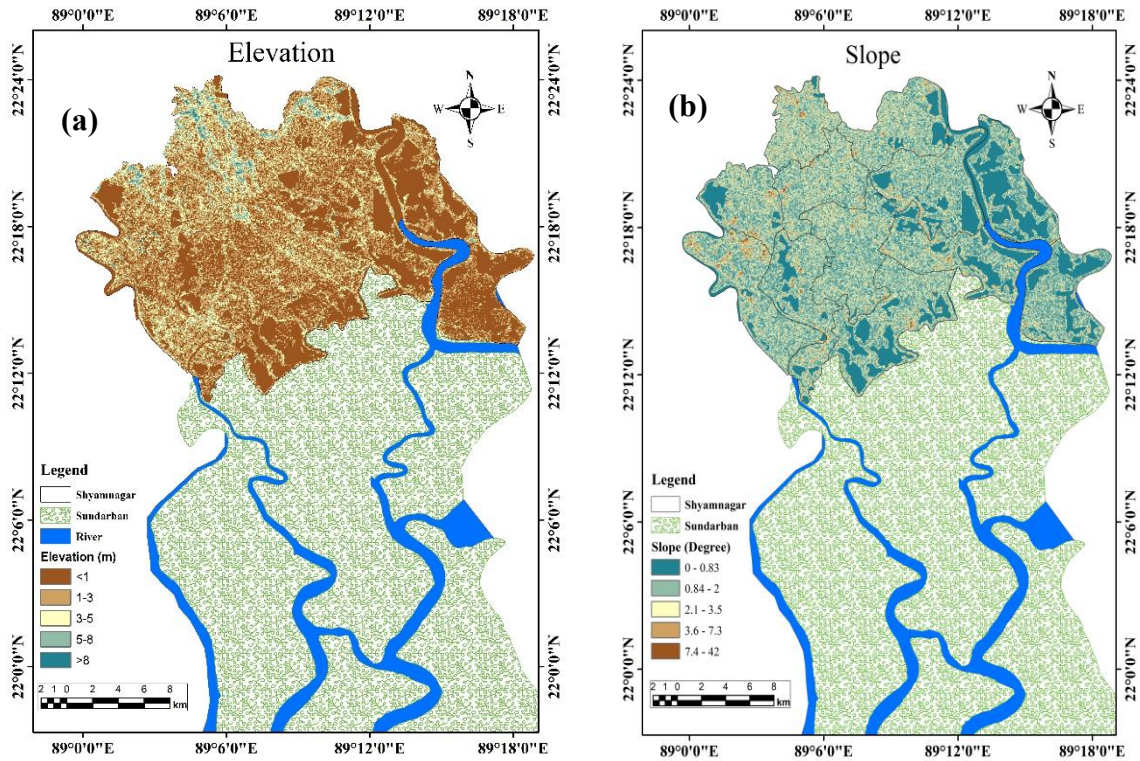


Figure 10: (a) Elevation (b) Slope

Proximity to coastline and cyclone track are important criteria in the cyclone vulnerability assessment. Life and properties near coasts and cyclone tracks are highly vulnerable to tropical cyclone impacts than those far from such areas. Google Earth Pro ruler was used to measure the distance of different sections of the study area from the coastline to produce the coastline proximity data. These data were then used to produce the proximity to coastline spatial layers (Fig. 11a). By comparison, the International Best Track Archive for Climate Stewardship (IBTrACS) data (www.ncdc.noaa.gov/ibtracs/) from 2000 to 2025 were used to prepare the proximity to the cyclone track spatial layer (Fig. 11b) (Knapp et al., 2010). In this process, about 6 spatial cyclone tracks were identified in the study area. Next, we performed the proximity analysis developing multi-ring buffers around the cyclone tracks using ArcGIS buffer tools to produce proximity to cyclone track spatial layer.

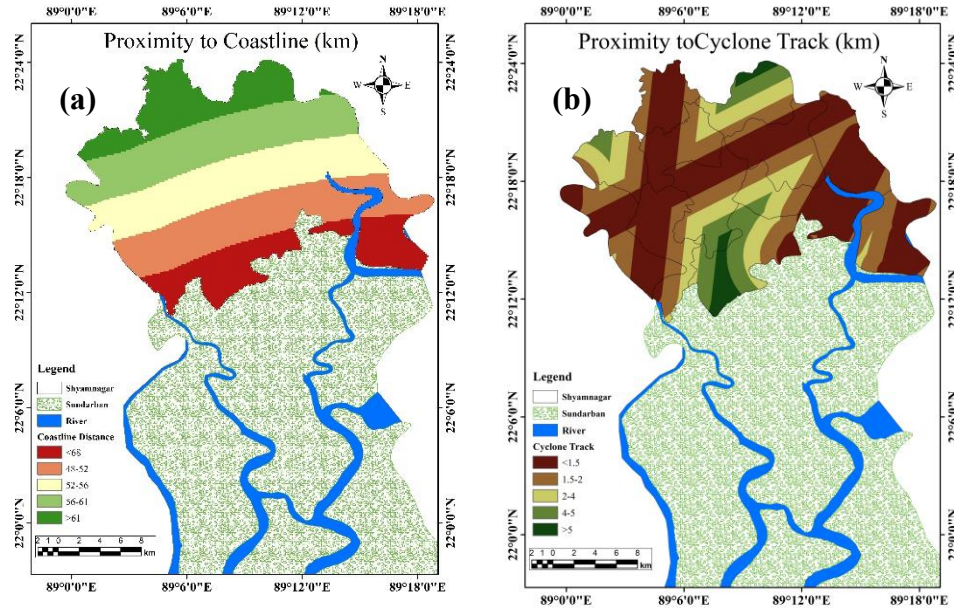


Figure 11: (a) Proximity to Coastline (b) Proximity to Cyclone Track

Land use and land cover (LULC) is a valuable exposure and indicator for tropical cyclone vulnerability assessment. In this study, seven Sentinel-2 imageries with a 10 m spatial resolution were used to generate the LULC map (Fig. 12a). The pre-processing of geometric, radiometric and atmospheric corrections of sentinel-2 images were performed using the Sen-2 Cor toolbox of Sentinel Application (SNAP) software. A hybrid classification scheme was adopted to classify six LULC categories, which include mixed Waterbody, Vegetation, Builtup, Agriculture, and Bareland. Initially, the probable classes were identified by applying an unsupervised clustering algorithm. Sample data were then selected for conducting supervised classification using a maximum likelihood algorithm (Kumar et al., 2013). All of these processes were performed using the GEE remote sensing software. The classification accuracy of the LULC map was determined using the random points acquired from the Google Earth imagery of the same periods. Having at least 50 points in each class, 505 total random points were generated using a stratified random sampling technique. Stratified random sampling is a probability sampling technique where the entire population is divided into different subgroups or strata, and then samples are collected randomly and proportionally from the different strata (A Ramezan et al., 2019). We followed the accuracy assessment explained in (Hoque et al., 2016; Jensen, 2005). The overall accuracy and kappa coefficient values were 92.08% and 90.94%, respectively.

The coastal population is rapidly increasing throughout the world, intensifying their exposure to tropical cyclones (Neumann et al., 2015). The spatial variation of population concentration and densities has a considerable influence on vulnerability levels (Poompavai and Ramalingam, 2013). In this study, the population density layer was prepared from the union profile of government of Bangladesh (Fig. 12b).

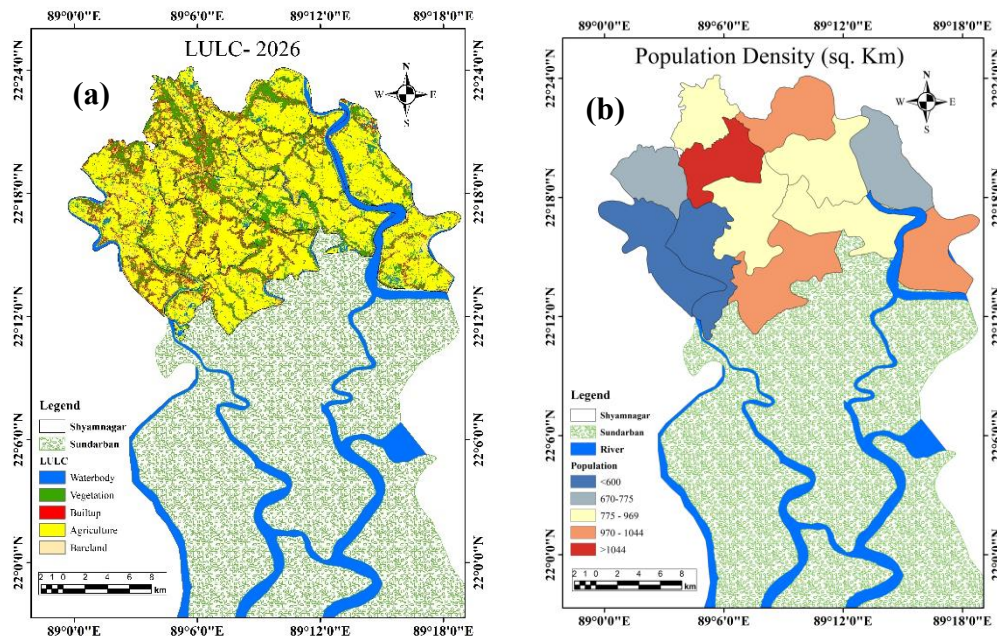


Figure 12: (a) Land Use and Land Cover (b) Population Density

3.5.2 Factors for Cyclone Hazards Evaluation

Hazard is any source of possible destruction, damage, harm or adverse effects on something or someone (Blaikie et al. 2014). Hazards are an extreme event that affects life, assets, and the environment (Rashid 2013). The hazardous effects of tropical cyclones are estimated through the record of historical data and present-day experience. The hazard probability is estimated based on the location, time, intensity and frequency of the hazardous events (Dewan 2013b). The factors which were selected to identify the potential cyclone risk for this study are precipitation/rainfall intensity, windspeeds, cyclone frequency, and Embankment Failure.

One of the catastrophic features of tropical cyclones is the storm surge, which can cause sudden flooding and significant damage to coastal communities. The storm Surge Composite Vulnerability Index (CVI) was calculated by integrating exposure (E), sensitivity (S), and resilience (R) (Fig.13). All indicators were normalized to a common scale (0–5) to ensure comparability. Exposure and sensitivity increase vulnerability, while resilience reduces it. Therefore, resilience was subtracted from the combined exposure and sensitivity.

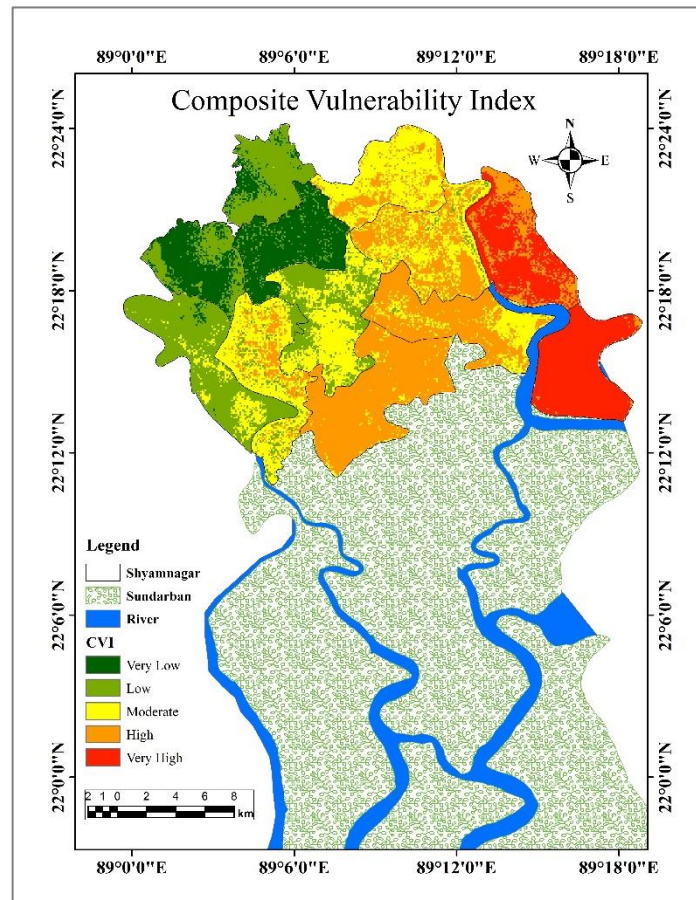


Figure 13: Storm Surge Composite Vulnerability Index

Cyclone wind speed is largely responsible for the destruction of infrastructures on the affected areas (Cardona et al., 2014). We prepared a cyclone wind speed spatial layer and considered the relevant data constraint following only the Cyclone Amphan (2020) wind speed spatial distribution data (Fig. 14a). A devastating tropical cyclone, Cyclone Amphan made landfall on 20th May 2020 in the southwestern coastal region of Bangladesh. With maximum sustained winds of 150 km/h, this cyclone caused considerable fatalities and damage.

Cyclone frequency influences the hazard intensity for a particular area because compared with occasional or rarely occurring hazards, frequent cyclone hazards present the great likelihood to affect an area (Poompavai and Ramalingam, 2013). In this study, we prepared the cyclone frequency spatial layer using the previous historical cyclone track data from 2000 to 2026 (Fig. 14b). From our field verifications, approximately 7 tropical cyclones have affected the study area over the last 68 years. The spatial analyst tool of ArcGIS (version 10.4) was used to convert the cyclone frequency data into a spatial layer.

Tropical cyclones cause extreme rainfall that ultimately intensifies flooding in the affected area. We were unable to collect spatial rainfall data linked to the landfall of tropical cyclones in the study area. Consequently, the daily precipitation data between 2000 and 2022 acquired from the Bangladesh Meteorological Department (BMD) were used to prepare the precipitation intensity map (Fig. 14c). Initially, we generated an annual precipitation intensity map by interpolating data.

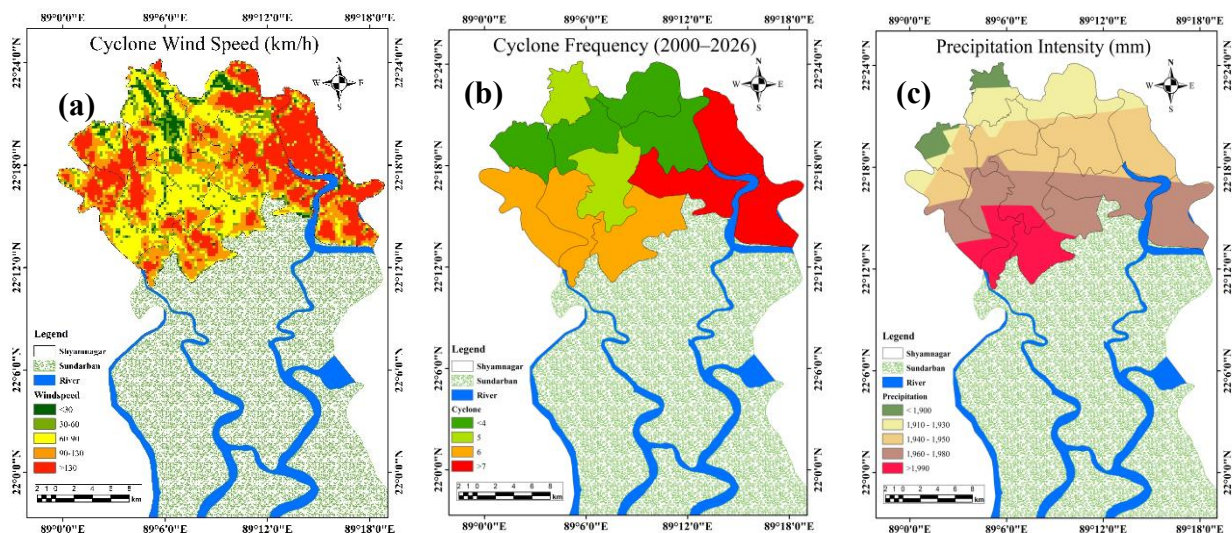


Fig. 14: (a) Cyclone Wind Speed (b) Cyclone Frequency (c) Precipitation Intensity

3.5.3. Criteria For Mitigation Capacity Mapping

Mitigation capacity indicates that key initiatives are planned and implemented to minimize the hazard impacts. We selected four mitigation capacity criteria (i.e., cyclone shelters, health facilities, coastal vegetation and cyclone warning systems) to assess the mitigation capacity in the study area. As an effective mitigation initiative, cyclone shelters provide emergency shelter to affected communities and protect them from the risk of storm surges during the cyclone event (Mallick and Rahman, 2013). Emergency health cares are likewise supported by the local health facilities prior to cyclone landfall. Cyclone shelter data were acquired from the Ministry of Disaster Management and Relief, whereas health facilities data was obtained from the local administrative office and verified in the field. We utilized the ‘Euclidian distance’ technique to prepare spatial layers of distance to cyclone shelter and health facilities (Fig. 15a, b).

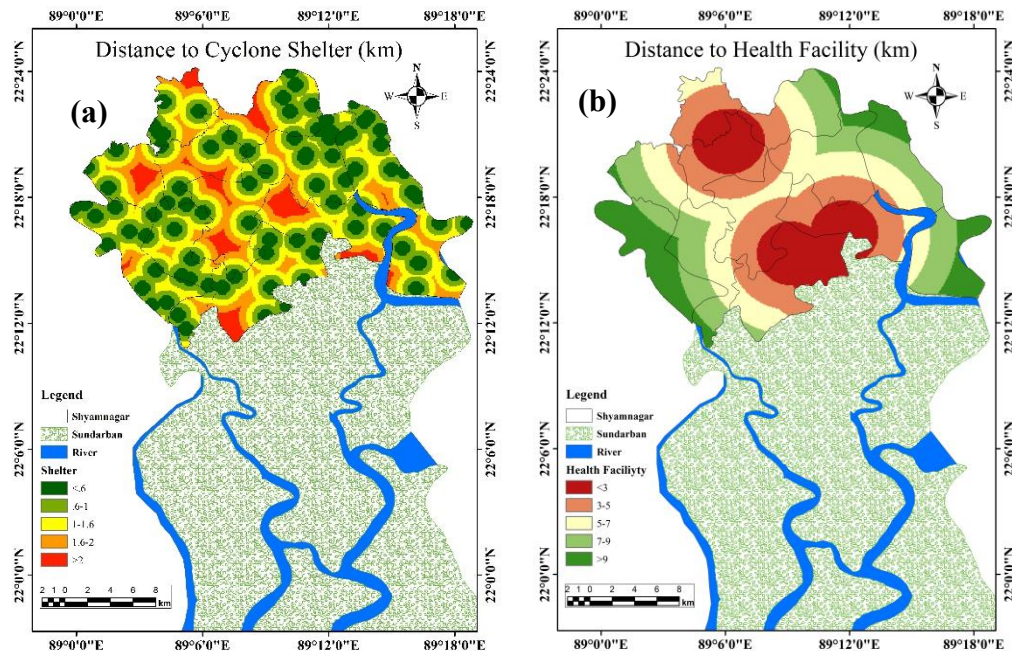


Figure 14: (a) Distance to Cyclone Shelter (b) Distance to Health Facility

Vegetation is considered protection during cyclone events to minimise the effects of wind and storm surges on coastal infrastructure and properties (Das and Vincent, 2009). In this study, The red and near-infrared (Band 4 and 5) of Landsat-8 imagery were used to map and identify different areas under vegetation cover (Fig. 16a). The NDVI was calculated using the formula as follows:

$$NDVI = \frac{(NIR-Red)}{(NIR+Red)} \dots\dots\dots 7$$

Where:

- **NIR** = Near Infrared band
- **Red** = Red band

In addition, an effective warning system plays a vital role to evacuate the people and belongings to reduce the impacts during cyclone events (Akhand, 2003). The warning system data including the information of equipment and processes were collected from the local administrative office and analyzed according to different parameters of the local warning system. The processed data was then verified in the field through discussions with local people and experts. Subsequently, the verified warning system information was categorized into three classes (i.e., High, moderate and Low) based on local administrative units (Fig. 16 b).

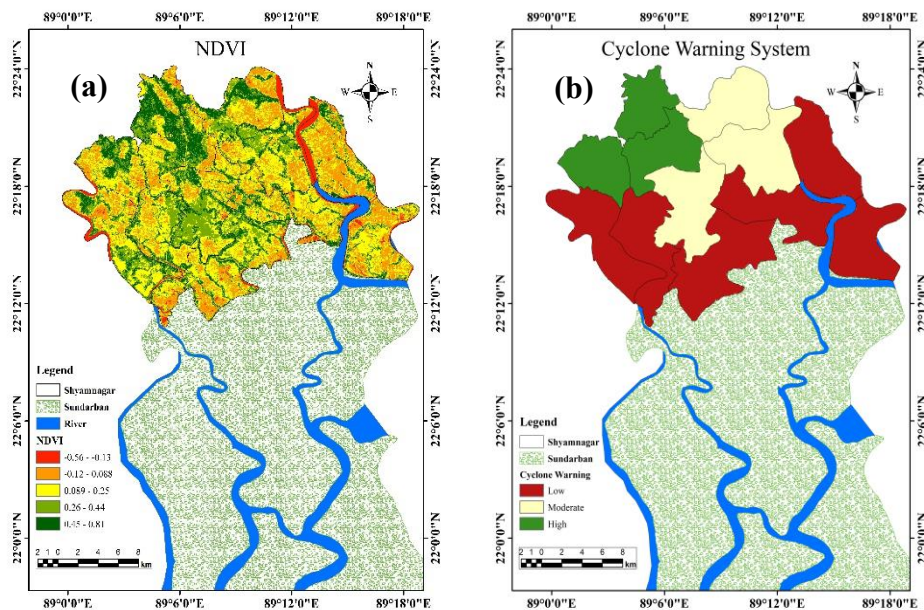


Figure 15: (a) NDVI (b) Cyclone Warning System

3.6. Alternative Ranking and Standardisation Of Criteria Layers

Risk ratings were assigned and ranked for each alternative of spatial criteria in the scale of 5-points, where 1 indicates very-low risk and 5 indicates very-high risk (Table 2). The alternatives were ranked considering the contribution to risk and AHP guidelines.

A 10-m pixel raster layer was created for each criterion, converting them from vector to raster to apply the weighted overlay technique. A standardisation process was performed using a linear scale transformation in Eq. 8 to bring the entire alternative ranking values into a common range from 0 to 1.

$$p = \frac{x - \min}{\max - \min} \dots\dots\dots 8$$

Exposure where p refers to standardised score; min and max indicate the minimum and maximum values of each dataset, respectively; and x means a value of the single cell in the dataset.

3.7. Weighting The Criteria Using AHP

In the present study, the AHP decision-making algorithm was adopted to weigh the criteria of the three risk components for developing a risk map. The pairwise comparison matrix was developed for each of the risk components with the help of four experts and a user, who followed the “scale of relative importance” developed by Saaty (2008). All of them had considerable experience in coastal research, cyclone factors and their influence on the study site. The total score of each risk component was. We calculated the consistency ratio (CR) to justify the consistency of comparison assigned by the experts and user. The comparisons are considered at the acceptable level if the CR value is equal or b0.1. The CR was calculated using Eq. (9):

$$CR = \text{Consistency} \frac{\text{Index}}{\text{Random}} \text{Index} \dots\dots\dots 9$$

where random index (RI) presents the randomly generated average consistency index, and the consistency index (CI) can be calculated as follows:

$$CI = \frac{\lambda_{\max} - n}{n - 1} \dots\dots\dots 10$$

where $\lambda_{\max}$ is the largest eigenvalue of the matrix and n denotes the matrix order (Malczewski, 2010). Table 3 outlines the criteria weights and consistency ratios calculated from the pairwise comparison.

3.8. Risk Assessment

On the basis of the weighted overlay technique, we developed the vulnerability and exposure, hazard and mitigation capacity indices incorporating their related criteria weights. Afterward, we produced vulnerability and exposure, hazard and mitigation capacity maps grouping the specific index values into five levels (i.e., very-low, low, moderate, high and very-high). Finally, an integrated risk index was generated using Eq. (1) following the raster calculator within the ArcGIS platform. A standardization process was then executed using Eq. (8) to bring the risk index values in the scale from 0 to 1. Thereafter, a risk map was produced to classify the risk index values into five categories of risk (i.e., very-low, low, moderate, high and very-high).

3.9. Risk Mapping Validation

The validation of spatial risk assessment results is challenging, as no specific validation approaches are widely established in the literature; therefore, a qualitative validation approach was adopted following Roy and Blaschke (2013). The study area in Shyamnagar was visited between October and December 2025 to obtain feedback from local people, experts, and policymakers (Fig. 16) regarding the generated risk assessment maps. Feedback was collected from all 384 respondents across the study area, ensuring representation of different socio-economic and environmental conditions, with a particular focus on historical cyclone frequency, perceived cyclone risk, health risk perception, and agricultural land transformation. In addition, in-depth field observation was conducted by identifying specific risk levels in different locations from the generated maps and comparing them with actual exposure to tropical cyclones, storm surge impacts, and associated environmental conditions, particularly in agricultural areas affected by salinity intrusion and waterlogging. Furthermore, historical tropical cyclone occurrences and their

impacts were documented through informal discussions with local people, including references to events such as Cyclone Aila and Cyclone Amphan.



Fig. 16: Visual survey on cyclone, health, and livelihood perception

Table. 2 Ranking of criteria alternatives following the contribution to tropical cyclone risks

Component	Criteria	Ranking (based on risk)				
		Very low (1)	Low (2)	Moderate (3)	High (4)	Very high (5)
Vulnerability and Exposure	Elevation (m)	>8	5-8	3-5	1-3	<1
	Slope (%)	0-0.83	0.84-2	2.1-3.5	3.6-7.3	7.4-42
	Proximity to coastline (km)	<48	48-52	52-56	56-61	>61
	Proximity to cyclone track (km)	<1.5	1.5-2	2-4	4-5	>5
	Land use and land cover	Waterbody	Bareland	Vegetation	Agriculture	Builtup
	Population Density	< 600	670-775	775-969	970-1044	>1044
Hazard	(CVI)	Very Low	Low	Moderate	High	Very High
	Cyclone frequency (number)		<4	5	6	7
	Precipitation (mm)	<1900	1900-1930	1930-1950	1950-1980	>1980
	Cyclone wind speed (km/h)	<30	30-60	60-90	90-130	>130
	Embankment Failure	0	1	2	3	>5
Mitigation capacity	NDVI	-0.56- -0.13	-0.12-0.08	0.09-0.25	0.26-0.44	0.45-0.81
	Cyclone warning system		High	Moderate	Low	
	Distance to cyclone shelter (Km)	<0.6	0.6-1	1-1.6	1.6-2	>2
	Distance to health facilities (Km)	<3	3-5	5-7	7-9	>9

Table. 3 Criteria weights and consistency ratios calculated from the pairwise comparison matrices.

Component	Criteria	Weight
Vulnerability and Exposure	Elevation (m)	0.18
	Slope (%)	0.12
	Proximity to coastline (km)	0.23
	Proximity to cyclone track (km)	0.25
	Land use and land cover	0.07
	Population Density	0.15
Hazard	(CVI)	0.36
	Cyclone frequency (number)	0.14
	Precipitation (mm)	0.11
	Cyclone wind speed (km/h)	0.20
	Embankment Failure	0.19
Mitigation capacity	NDVI	0.18
	Cyclone warning system	0.30
	Distance to cyclone shelter (Km)	0.44
	Distance to health facilities (Km)	0.08
Consistency ratio: 0.03		

Chapter four: Result

4.1. Spatial Modeling Of Storm Surge Flood Inundation

We used DEM to simulate the effects of different flood inundation levels ranging from 0.5 m to 5 m (Fig. 17). A direct relationship between flood water level and inundation extent was observed (Fig. 4), where the inundated area increased progressively with rising water levels. At 0.5 m inundation, about 31% of the total study area is submerged, primarily affecting agricultural land (18.6%), followed by vegetation (4.65%), built-up areas (3.1%), and bare land (1.55%). When the inundation level reaches 1 m, nearly 47% of the area becomes inundated, with agricultural land (28.2%) experiencing the highest impact, along with noticeable submergence of vegetation (7.05%) and settlements (4.7%).

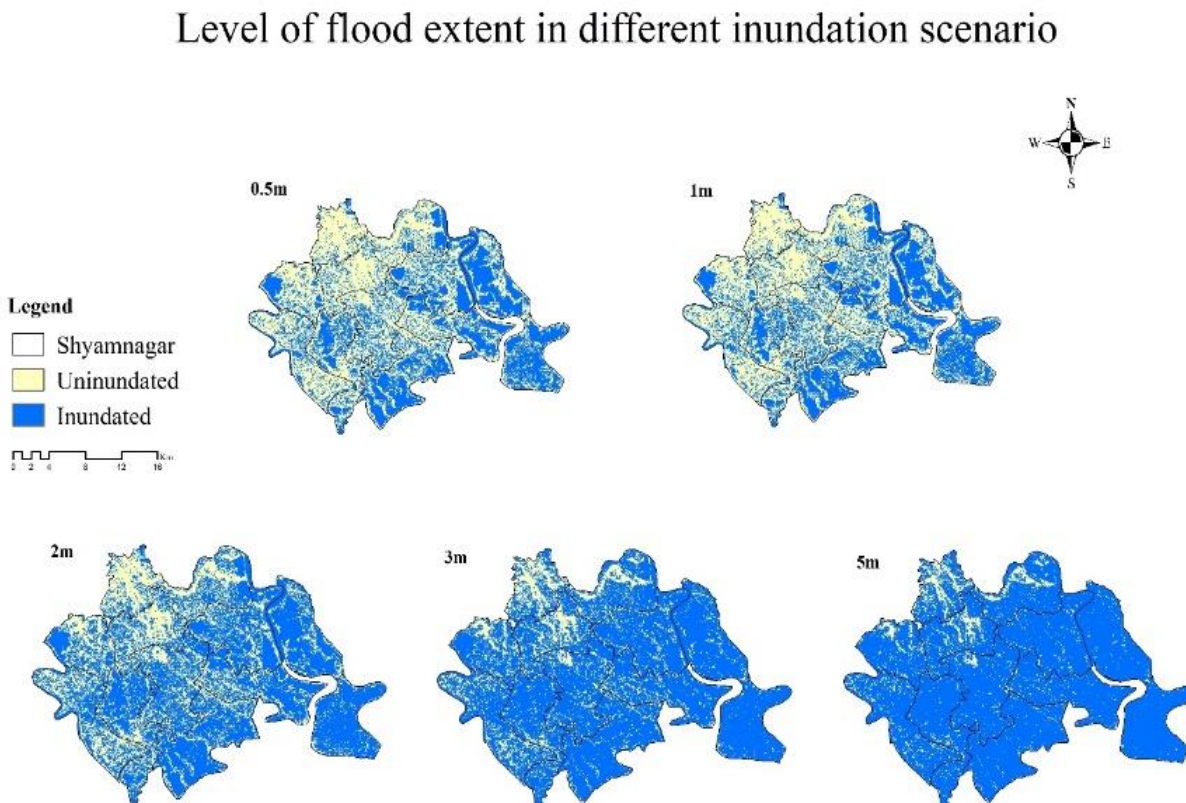


Figure 16. Level of flood extent in different inundation scenarios (0.5-3, 5 m) in the study area.

With a further increase to 2 m inundation, approximately 65% of the study area is submerged, where agricultural land (39%) and vegetation (9.75%) are extensively affected, and about 6.5% of built-up areas are inundated, indicating significant socio-economic disruption. The impact on all LULC categories intensifies with increasing flood levels, as observed at 3 m inundation, where nearly 81% of the total area is submerged, including 48.6% agricultural land, 12.15% vegetation, and 8.1% built-up areas. This demonstrates a substantial escalation in both environmental and settlement vulnerability. At the highest simulated flood level of 5 m, around 91% of the study area becomes inundated, with more than half of the agricultural land (54.6%) affected, along with significant portions of vegetation (13.65%), built-up areas (9.1%), and bare land (4.55%).

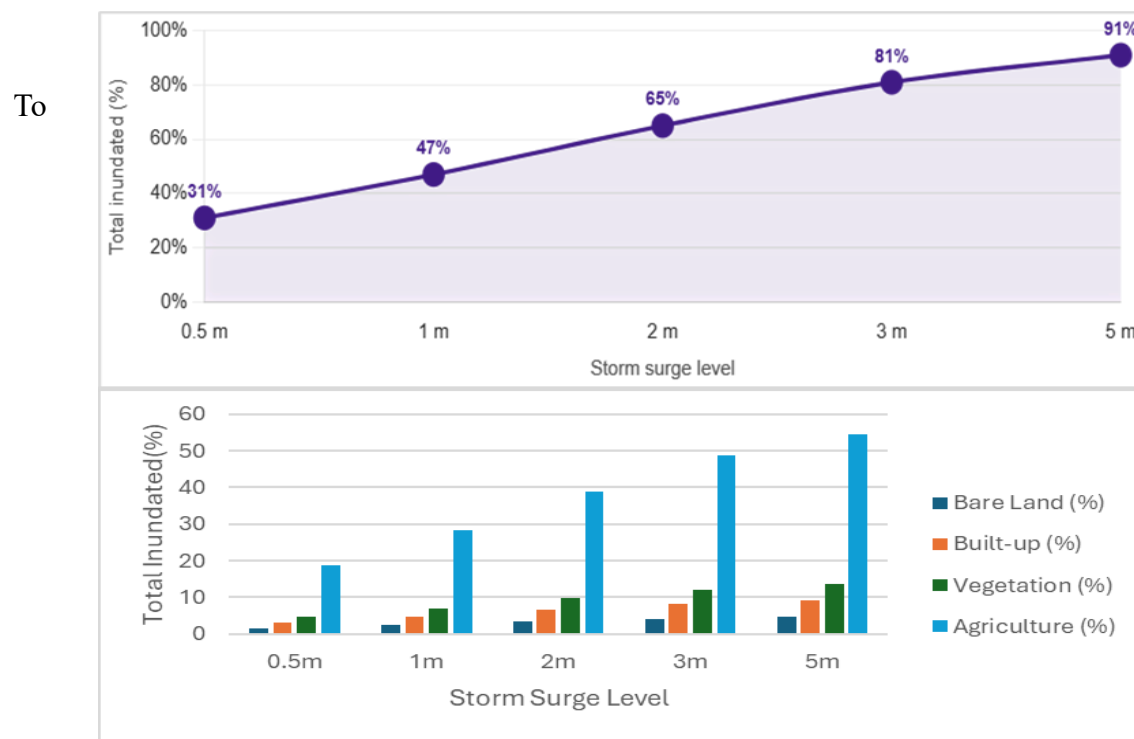


Figure 17. Level of flood inundation (0.5–5 m) and affected area under various land use/land cover classes in Shyamnagar.

Storm surge inundation increases sharply with rising water levels across all unions. At 0.5 m, inundation remains relatively low (≈ 10 – 55%), but it expands significantly at higher depths,

reaching above 90% in most unions at 5 m. Gabura, Padma Pukur, and Munshiganj show the highest vulnerability, while Bhurulia remains comparatively less affected. Overall, the results indicate severe flood risk escalation with increasing storm surge depth in Shyamnagar's coastal unions. Further understand spatial heterogeneity, union-wise inundation analysis reveals strong variability across the study area (Fig. 19)

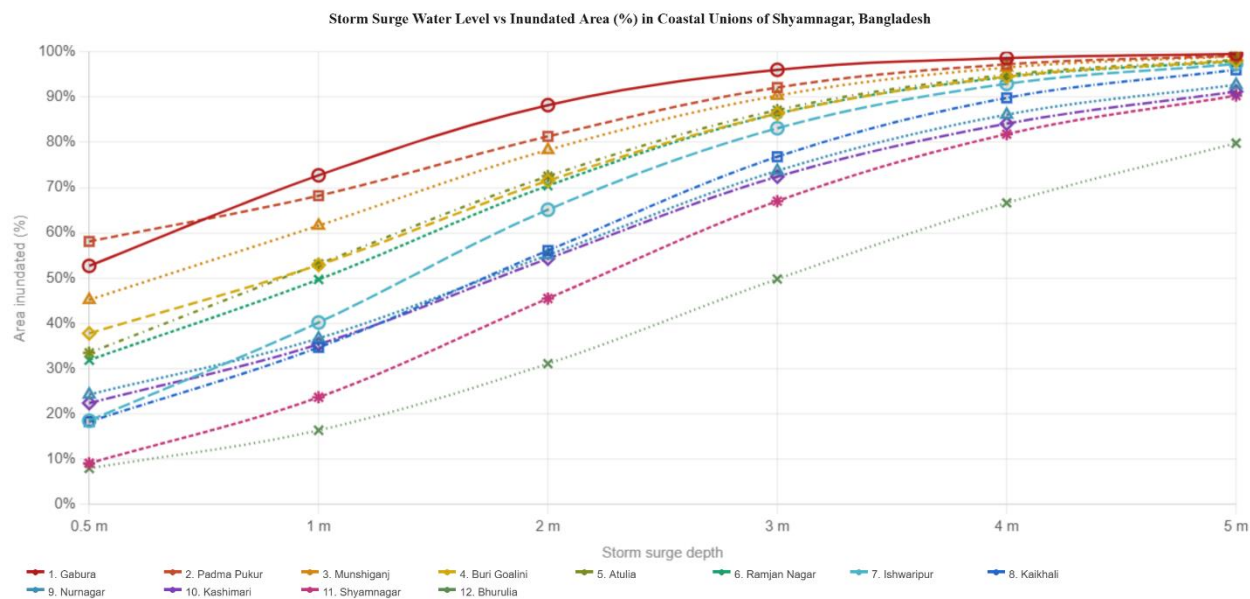


Figure 18: Union Based Storm Surge Water Level vs Inundated Area (%)

4.2. Flood Inundation Risk

Flood inundation risk in the Shyamnagar region was assessed using Eq. (2) based on union-level analysis (Fig. 5). The results indicate that Gabura Union, Poddopukur Union, and Burigoalini Union fall under the very high flood inundation risk category, as a substantial portion of these areas becomes submerged even at lower inundation levels. These unions are located in the south and south-eastern part of the study area, characterized by low-lying topography and high exposure to coastal flooding.

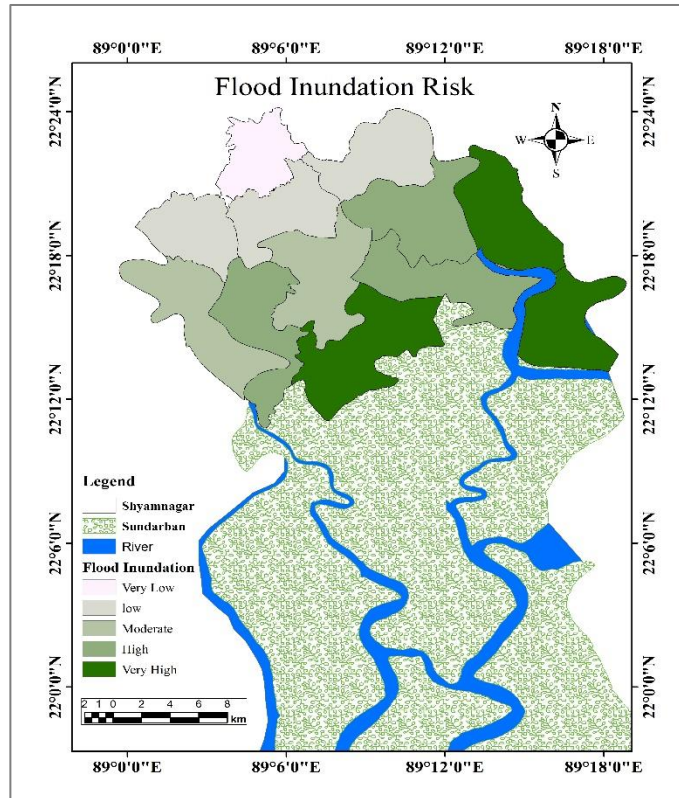


Figure 19: Flood Inundation map

This finding is further supported by the impact of Cyclone Amphan, during which significant inundation was observed in Gabura and Poddopukur unions, as evidenced by the prepared inundation map (Fig. 21).

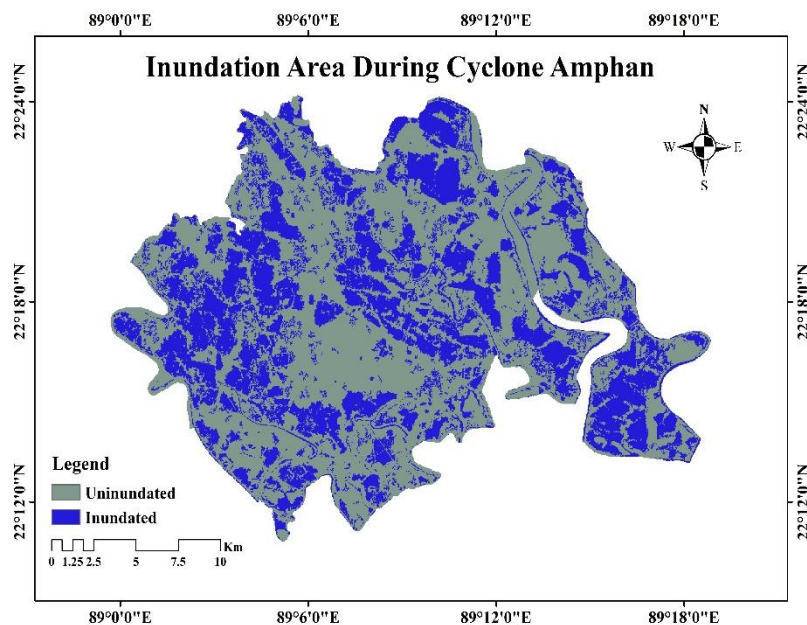


Figure 20: Assessment of Inundated Area During Cyclone Amphan

In contrast, Bhurulia Union exhibits the lowest flood inundation risk, likely due to its relatively higher elevation and reduced hydrological connectivity to coastal flood sources. Overall, the spatial distribution of flood risk clearly shows that the southern and south-eastern coastal unions are the most vulnerable, while inland areas experience comparatively lower risk.

4.3. Exposure, Sensitivity, Resilience and CVI of Storm Surge Flood

Out of the total analyzed unions, the distribution of exposure to storm surge flood shows that a small proportion of unions fall under the extreme categories, while most are distributed across low to high classes. Specifically, very low exposure is observed in 1 union, followed by low exposure in 2 unions, medium exposure in 3 unions, and high exposure in 4 unions, whereas very high exposure is identified in 2 unions (Table 4).

Table. 4 Categorical Distribution of Exposure, Sensitivity, Resilience, and CVI Across Study Area Unions

Category	Exposure (No. of Union)	Sensitivity (No. of Union)	Resilience (No. of Union)	CVI (No. of Union)
Very low	1	0	1	1
Low	2	1	1	3
Medium	3	4	7	2
High	4	5	1	3
Very High	2	1	2	2

This indicates that although a limited number of unions experience very high exposure, a considerable proportion remains under medium to high exposure conditions. The spatial pattern suggests that exposure is primarily influenced by geomorphological and hydrodynamic conditions such as low elevation, proximity to tidal channels, and susceptibility to storm surge inundation, which collectively increase exposure intensity in coastal-adjacent unions. (Fig. 22)

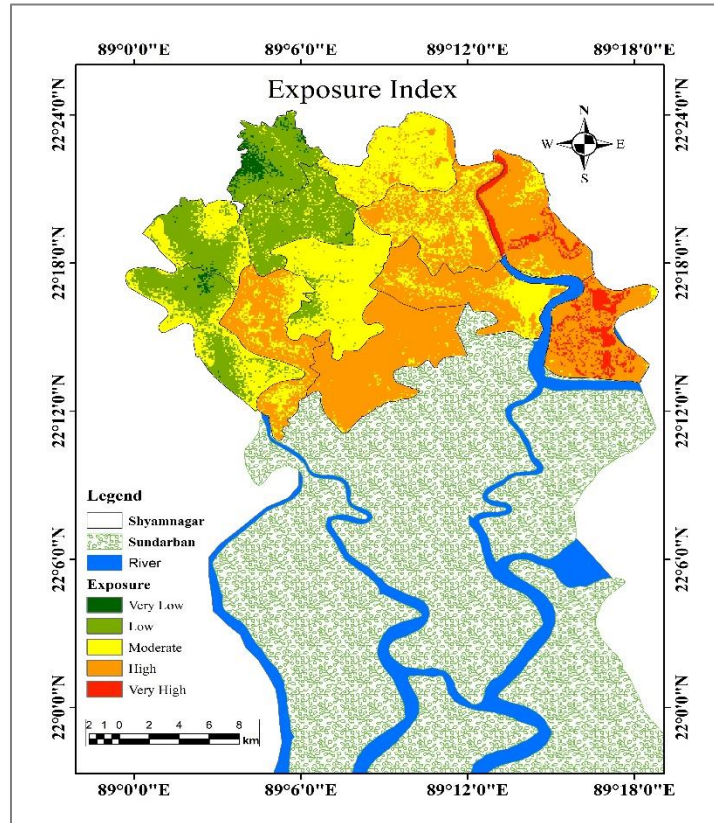


Figure 21: Exposure Map

The sensitivity analysis indicates that the majority of unions fall within moderate to high sensitivity categories. Specifically, 5 unions are classified as highly sensitive, followed by 4 unions with medium sensitivity, while only 1 union falls under low sensitivity and 1 under very high sensitivity; no union is identified as very low sensitive. This distribution highlights that a significant portion of the study area is inherently vulnerable to storm surge impacts. The elevated sensitivity is primarily driven by high population density, concentrated settlements, dependence on agriculture, and fragile housing conditions. Additionally, inadequate access to healthcare, safe water, and sanitation further intensifies the susceptibility of these communities. Consequently, even lower levels of hazard exposure can lead to substantial socio-economic and health-related impacts in the study area (**Fig. 23**).

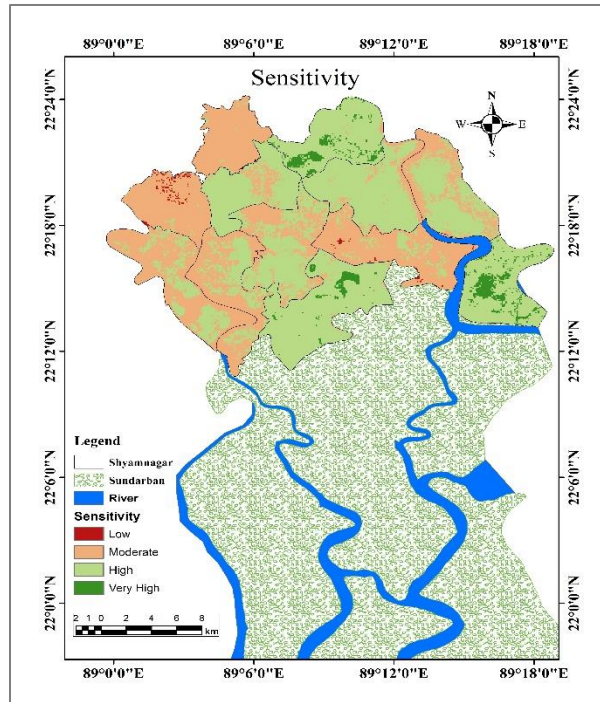


Figure 22: Sensitivity Map

The resilience capacity across the unions demonstrates a relatively mixed pattern, with 1 union each in very low and low resilience categories, 7 unions in the medium category, 1 union in the high category, and 2 unions in the very high resilience classes. This suggests that the majority of unions possess moderate resilience, while only a few exhibit strong adaptive and coping capacities. Variations in resilience are mainly driven by differences in infrastructure development, livelihood diversification, literacy rate, access to financial services, and communication facilities. **(Fig. 24).** These spatial differences in resilience indicate unequal capacity to recover from cyclone-induced hazards across the study area. Unions with higher resilience are likely to experience faster post-disaster recovery and lower long-term damage. Conversely, low-resilience unions remain more exposed to repeated climate shocks and require targeted intervention.

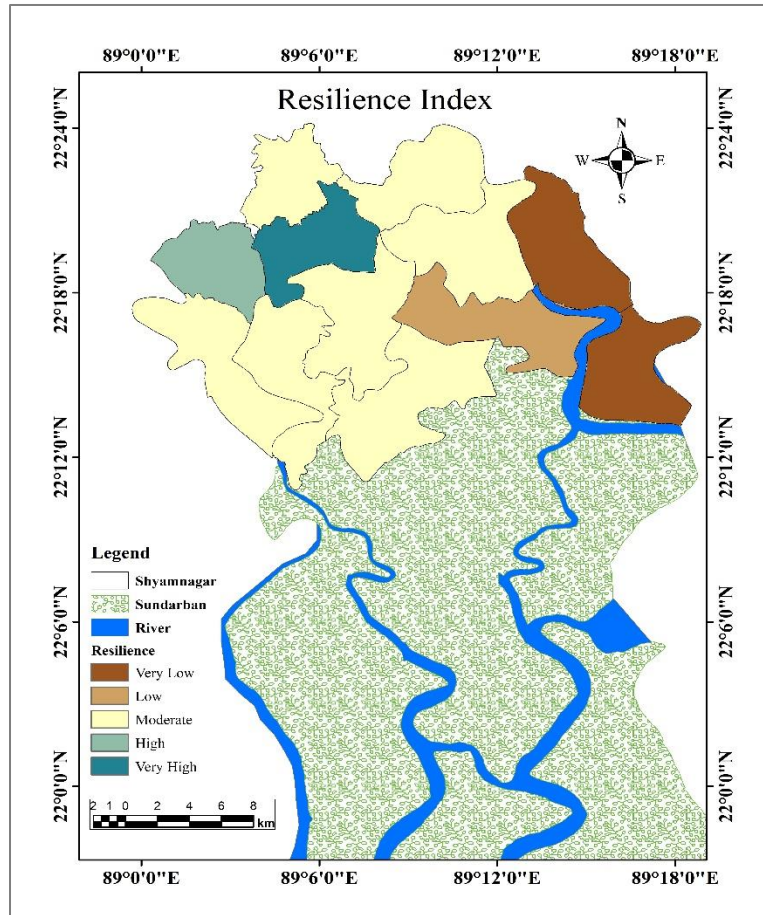


Figure 23: Resilience Index

Finally, the Composite Vulnerability Index (CVI) indicates that 1 union falls under very low vulnerability, 3 unions under low vulnerability, 2 unions under medium vulnerability, 3 unions under high vulnerability, and 2 unions under very high vulnerability. This distribution highlights a heterogeneous vulnerability pattern across the study area, where a significant number of unions still experience high to very high vulnerability conditions. Overall, the CVI results suggest that despite the presence of some relatively resilient areas, a considerable portion of the region remains highly susceptible to storm surge flood impacts due to the combined influence of exposure, sensitivity, and limited resilience. (Fig. 25)

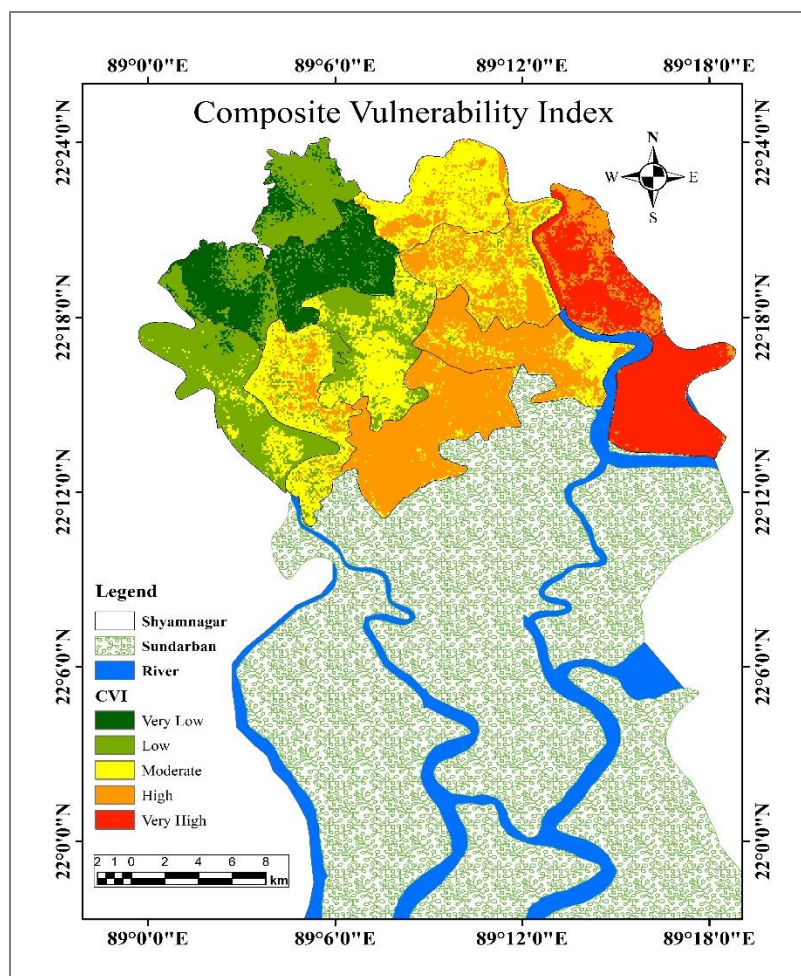


Figure 24: Composite Vulnerability Index

4.4. Vulnerability, Hazard and Mitigation mapping for Risk Map

The spatial analysis of cyclone vulnerability across the 12 unions of Shyamnagar Upazila reveals a predominantly high-to-very-high vulnerability profile. The vulnerability map was classified into five categories, Very Low, Low, Moderate, High, and Very High, based on composite socio-physical indicators. The dominant zone across the upazila is High vulnerability, covering approximately 45–50% of the total area, concentrated across the central and southwestern unions such as Burigoalini, Gabura, Padmapukur, and Ramjanpur. Very High vulnerability accounts for roughly 15–20% of the area, clustered particularly in the deep-western and northwestern unions of Ishwaripur and Atulia, where settlement density and exposure to tidal flooding amplify risk. Moderate vulnerability covers approximately 20–25% of the upazila, predominantly in the northern unions including Nurnagar, Kashipur, and portions of Bhurulia. Low and Very Low vulnerability zones are spatially restricted to small pockets in the far north, representing only about

5–8% of the study area combined, largely corresponding to areas with denser vegetation and slightly elevated landforms (Fig. 26).

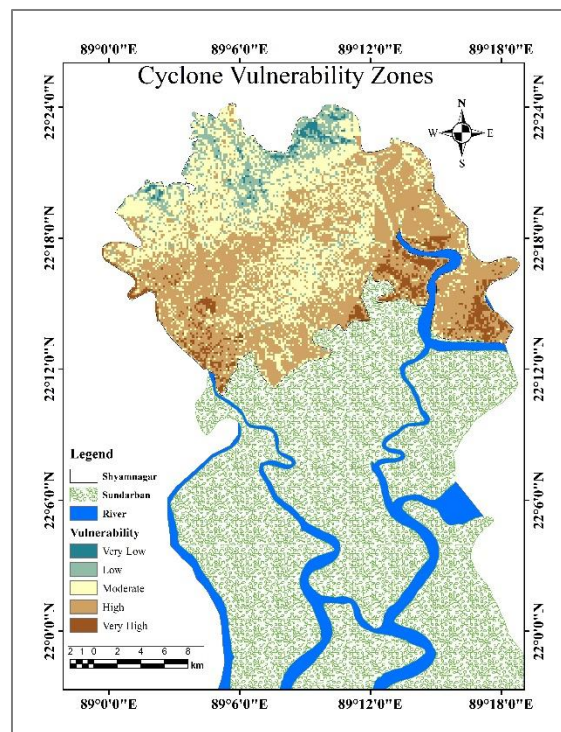


Figure 25: Cyclone Vulnerable Zones

The hazard assessment, derived from wind exposure, storm surge inundation probability, and proximity to tidal channels, shows a pronounced east-to-west gradient across Shyamnagar Upazila. Very High hazard conditions dominate the eastern unions, particularly Shyamnagar Sadar, Kashmiri, and parts of Nurnagar, covering an estimated 20–25% of the upazila's settled area. These areas lie in immediate proximity to the Kholpetua and Kobadak rivers and face maximum exposure to Bay of Bengal-originating cyclones making landfall along the south-eastern coast. High hazard zones account for a further 15–20% of the area, spreading through central unions such as Ramjanpur and Mathbaria. Moderate hazard is the single largest hazard class, representing approximately 25–30% of the settled upazila, spanning a broad central belt. The Low and Very Low hazard zones, occupying around 15–20% combined, are concentrated in the northwestern sector, particularly the Atulia and Bhurulia unions, where distance from the coast and the partial buffering effect of the Sundarban mangrove system attenuate hazard intensities (Fig.27).

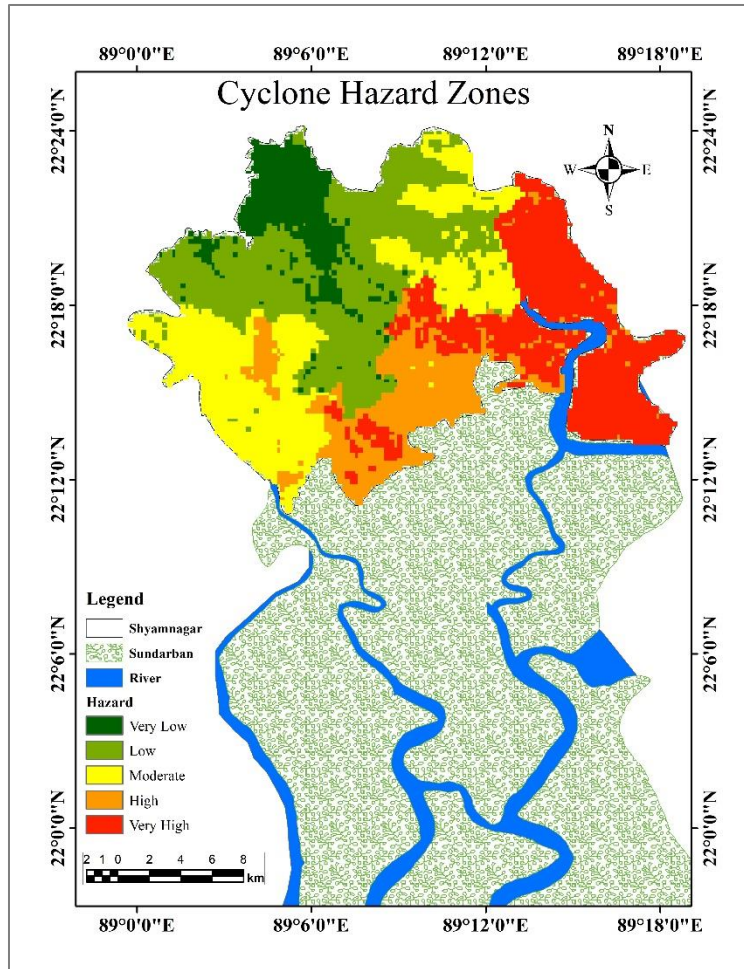


Figure 26: Cyclone Hazard Zone

The mitigation capacity map presents the inverse of vulnerability, capturing the ability of communities and ecosystems to reduce cyclone impacts through infrastructure, early warning systems, embankment density, shelter availability, and mangrove cover. The analysis reveals that Moderate mitigation capacity is the dominant class, covering approximately 35–40% of the upazila, distributed across the central unions. Low mitigation capacity is prevalent across a large contiguous zone in the central-west and south, accounting for roughly 25–30% of the area, and corresponding spatially to the same unions that exhibit high vulnerability, a reinforcing pattern that underscores the compound risk faced by Gabura, Burigoalini, and Padmapukur. Very Low mitigation capacity covers approximately 15–20% of the area, predominantly in the southern settled fringes adjacent to the Sundarban. In contrast, High and Very High mitigation capacity zones are spatially restricted to the northern and northeastern unions, particularly near Shyamnagar Sadar and portions of Nurnagar, where road access, embankment quality, and cyclone shelter

availability are comparatively better; together these classes account for approximately 10–15% of the total area (Fig.28).

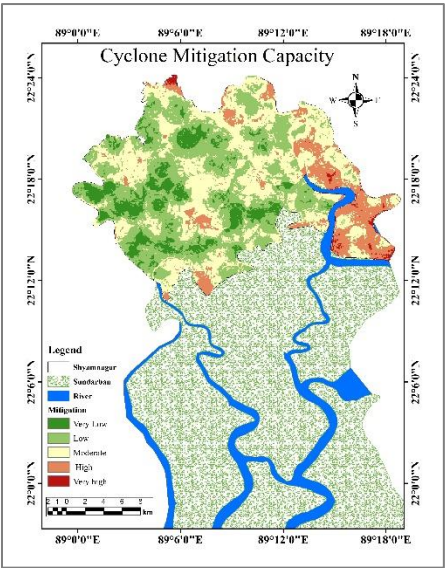


Figure 27: Cyclone Mitigation Capacity in Shyamnagar

The spatial distribution of vulnerability, hazard, and mitigation capacity shows notable variation across the study area (Table 5). Vulnerability is dominated by the high class (48%) followed by very high (22%), indicating widespread susceptibility to cyclone impacts, while hazard is also concentrated in moderate to very high classes, covering the majority of the area and reflecting significant exposure to storm surge inundation. In contrast, mitigation capacity is relatively moderate, with the highest proportion in the moderate class (38%), followed by high (27%) and very high (20%), though low-capacity areas still persist. Overall, the results reveal that high vulnerability and hazard conditions are not sufficiently balanced by strong mitigation capacity, indicating persistent risk across the study area.

Table. 5 Approximate Percentage Area by Risk Class

Class	Vulnerability (% Area)	Hazard (% Area)	Mitigation Capacity (% Area)
Very Low	3%	8%	5%
Low	5%	12%	10%
Moderate	22%	28%	38%
High	48%	30%	27%
Very High	22%	22%	20%

4.5. Risk Mapping

The spatial distribution of tropical cyclone risk in Shyamnagar Upazila shows clear heterogeneity among the unions (Fig. 29). The analysis reveals that very high-risk areas constitute 9% (6.9 km²; 2 unions), while high-risk areas cover 27% (20.7 km²; 3 unions). These high and very high-risk unions are mainly located in the southern and south-western coastal fringe of the study area, including Gabura and Padmapukur (very high risk) and Burigoalini, Munshiganj and Kaikhali (high risk), where exposure to storm surge flooding is extremely high and resilience capacity is relatively low. Specifically, the very high-risk unions such as Gabura and Padmapukur are located in the extreme coastal and riverine interface zone, where frequent tidal inundation, low elevation, and limited livelihood diversification intensify cyclone impacts. The high-risk unions including Burigoalini, Munshiganj, Kaikhali, and part of Kashimari are distributed along the western and south-western coastal belt, where proximity to tidal rivers, cyclone tracks, and weak protective infrastructure further increases disaster risk conditions. Moderate risk covers 18% (13.8 km²; 2 unions) of the study area, mainly including Atulia, Ramjannagar and Ishwaripur, which represent transitional zones where exposure is moderate and some resilience support exists due to improved infrastructure and livelihood diversity compared to the extreme coastal fringe. In contrast, low-risk and very low-risk zones together account for 46% (35.4 km²; 5 unions). These areas are mostly located in the northern and north-eastern inland parts of Shyamnagar Upazila, including Bhurulia, Nurnagar, Shyamnagar Sadar and parts of Ramjannagar, where relatively higher elevation, reduced direct tidal influence, and better access to services contribute to lower risk levels. The very low-risk condition is mainly observed in Shyamnagar Sadar and inland sections of Nurnagar, where exposure to storm surge is minimal compared to coastal unions.

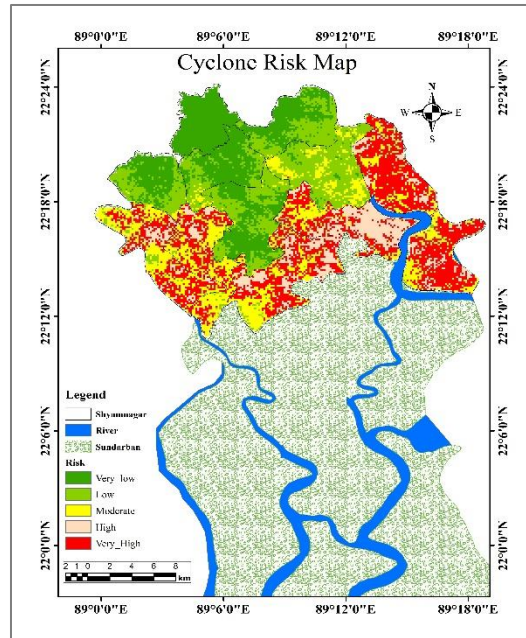


Figure 28: Cyclone Risk Map

4.6. Health Risk Perception

The spatial distribution of climate-related health impacts across the study unions shows clear heterogeneity (Figure 6). Overall, skin disease is the most prevalent health issue across all unions, followed by high blood pressure, scabies, and gastric problems. Among the studied areas, Gabura exhibits the highest burden of health impacts, with skin disease reaching 78%, high blood pressure 50%, scabies 45%, and gastric problems 40%. Similarly, Munshiganj also shows elevated health risks, particularly for skin disease (72%) and high blood pressure (55%), indicating a relatively high vulnerability compared to other unions. Moderately high prevalence is observed in Padma Pukur, where high blood pressure (58%) and scabies (52%) are notably high, alongside skin disease (60%) and gastric issues (28%). Buri Goalini and Kashimari also show substantial skin disease prevalence (65% and 68%, respectively), though gastric disease remains comparatively lower. In contrast, lower health impacts are recorded in Shyamnagar and Nurnagar, where all health indicators remain comparatively low, with skin disease at 33% and 42%, respectively, and minimal gastric prevalence (9–12%). Bhurulia also shows consistently low values across all health conditions.

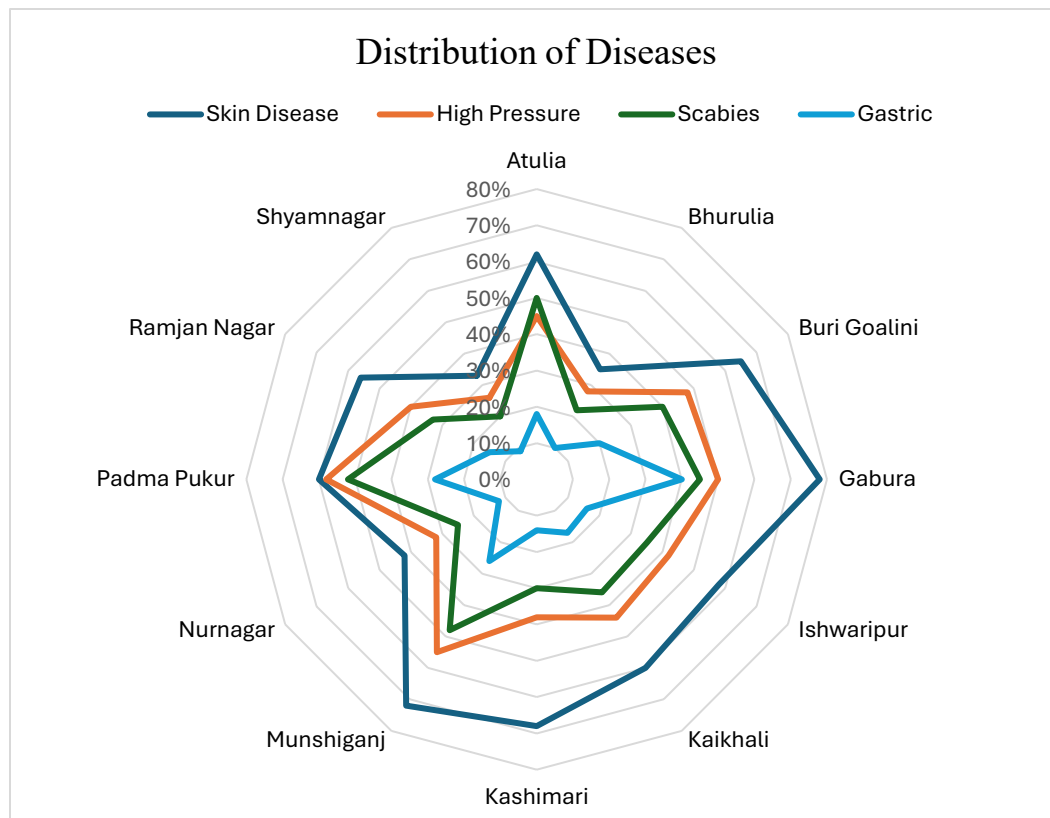


Figure 29: Health Impacts Across Unions

4.7. Agriculture Land use analysis

The conversion rate across the selected unions shows notable spatial variation, indicating clear differences in the level of adoption or transition among the study areas (Figure X). Overall, the conversion rate ranges from a minimum of 28% to a maximum of 68%, reflecting substantial heterogeneity among unions. The highest conversion rate is observed in Gabura with 68%, followed by relatively high rates in Kaikhali (60%) and Ramjan Nagar (56.4%). Munshiganj and Padma Pukur also show moderately high conversion rates of 54% and 51%, respectively, suggesting stronger transformation processes in these unions. In contrast, the lowest conversion rates are recorded in Bhurulia (28%), Shyamnagar (29%), and Nurnagar (33%), indicating relatively slower conversion dynamics in these areas. Kashimari also shows a comparatively low rate of 40.2%, highlighting moderate to low conversion performance.

UNION-LEVEL ANALYSIS

Agricultural land use conversion to fisheries

Conversion rate by union — ranked high to low

Unions surveyed	Highest rate	Average rate	Lowest rate
12	68.0%	47.6%	28.0%

● High conversion (≥55%) ● Moderate (40–54%) ● Low (<40%)

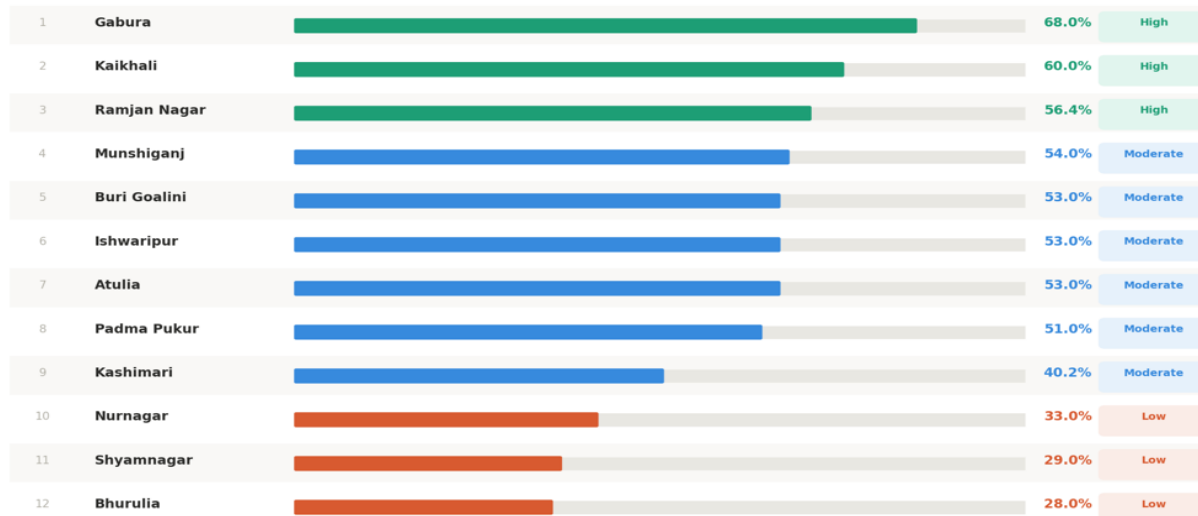


Figure 30: Union-level Variation in Conversion Rate

Chapter Five: Discussion

The integrated assessment of storm surge inundation, vulnerability, health outcomes, and land-use transformation in Shyamnagar Upazila reveals a strongly structured and spatially coherent pattern of coastal risk. The results demonstrate a clear amplification effect, where increasing inundation depth progressively intensifies exposure of both ecological and socio-economic systems. At the highest simulated scenario, inundation expands to approximately 91% of the study area, with agricultural land consistently representing the most affected land-use class. This dominance highlights the agrarian foundation of the local economy and its inherent sensitivity to even moderate storm surge events, consistent with findings from the Sundarban Biosphere Reserve where post-inundation salinity intrusion severely reduces cropland productivity (Sahana & Sajjad, 2019).

A pronounced north–south risk gradient emerges across all analytical layers. Southern coastal unions such as Gabura, Padma Pukur, and Burigoalini consistently exhibit the highest inundation exposure, composite vulnerability, and health burden. In contrast, relatively elevated inland areas such as Bhurulia demonstrate consistently lower risk profiles. This spatial differentiation is primarily controlled by topographic elevation, proximity to tidal channels, and hydrological connectivity, which are widely recognized as dominant drivers of flood risk in deltaic environments (Hazra et al., 2017, as cited in Sahana & Sajjad, 2019; Myers et al., 2008, as cited in Sahana & Sajjad, 2019). The Composite Vulnerability Index further confirms that vulnerability is not solely hazard-driven but strongly shaped by socio-economic sensitivity and uneven resilience capacity. Unions with high population dependence on agriculture, limited livelihood diversification, and weaker infrastructure access exhibit significantly higher vulnerability scores. This pattern aligns with broader vulnerability frameworks in coastal deltas, where socio-economic constraints are identified as primary determinants of disaster impact severity (Fothergill and Peek, 2004, as cited in Sahana & Sajjad, 2019; Tessler et al., 2015, as cited in Sahana & Sajjad, 2019).

Health outcomes show a direct and consistent spatial correspondence with inundation and vulnerability patterns. Higher prevalence of skin disease, hypertension, scabies, and gastric disorders is concentrated in highly exposed southern unions, particularly Gabura and adjacent areas, reflecting prolonged exposure to saline water intrusion, sanitation disruption, and environmental degradation. These findings align with established evidence that storm surge

flooding generates long-term public health burdens through contaminated water systems and post-disaster environmental stress (Sahana & Sajjad, 2019).

Land-use transformation patterns further reinforce cumulative climate stress dynamics. Higher agricultural conversion rates in exposed unions such as Gabura (68%) indicate a forced adaptation process driven by repeated salinity intrusion and declining agricultural viability rather than voluntary economic transition. This is consistent with documented trends in the Bengal delta, where storm surge-induced soil salinization progressively renders traditional rice cultivation unsustainable (Sahana & Sajjad, 2019).

The findings highlight a cascading risk structure in Shyamnagar Upazila, where physical exposure, socio-economic vulnerability, health impacts, and livelihood disruption are tightly interconnected. The spatial convergence of these stressors in southern coastal unions underscores the need for geographically targeted adaptation strategies rather than uniform interventions. Priority should focus on embankment strengthening, infrastructure development, and livelihood diversification in high-risk zones, consistent with coastal resilience strategies proposed for the Sundarban region (Dasgupta et al., 2011; Danda, 2007; HDR, 2009, as cited in Sahana & Sajjad, 2019).

Chapter Six: Conclusion

6.1 Conclusion

This study demonstrates that cyclone risk in Shyamnagar Upazila is not uniformly distributed but follows a pronounced and spatially structured north–south gradient, where southern coastal unions, particularly Gabura, Padma Pukur, and Buri Goalini, face a compounding cycle of severe storm surge exposure, socio-economic sensitivity, and critically limited adaptive capacity, as evidenced by the finding that 44.3% of the upazila falls under high to very high risk. The DEM-based inundation modeling, validated against Cyclone Amphan observations, identifies a critical 2-metre surge threshold beyond which flooding transitions from environmental damage to acute socio-economic disruption, while the AHP-weighted Composite Vulnerability Index confirms that the same southern unions bearing the greatest physical hazard also report the highest prevalence of climate-sensitive diseases, including skin disease (58%), scabies (36%), and hypertension (30%), and the most severe agricultural land conversion, with 68% of Gabura's farmland already shifted to aquaculture, compared to a 46.7% upazila average, reflecting stress-driven livelihood collapse rather than voluntary diversification. These converging spatial patterns carry clear and urgent policy implications: cyclone shelter expansion, coastal embankment reinforcement, and health infrastructure investment must be prioritized in high-risk southern unions, while salinity-tolerant agricultural programs and soil rehabilitation are essential to arrest further productive land loss before it becomes irreversible. The Storm Surge Composite Vulnerability Index framework developed here is methodologically transferable to other vulnerable coastal upazilas across Bangladesh, offering a scalable tool for national-level disaster risk planning. As climate change intensifies cyclone frequency and sea-level rise accelerates, the spatially explicit evidence produced by this study makes one conclusion unavoidable: the window for proactive intervention in communities like Gabura is narrowing, and delay is no longer an option.

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